KTH Industrial Engineering and Management

Resource energy efficiency measures for retail sector in Azerbaijan

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ABSTRACT

The thesis work is part of a larger project financed by the European Bank for Reconstruction and Development (EBRD) and awarded by RINA Consulting S.p.A.. Azeri Retail will receive the loan from EBRD in order to refurbish its six new acquisitions in Baku, Azerbaijan. Azeri Retail considers the current Fresco format store a good technological and structural level, therefore Azeri Retail would like to replicate this state also for the new acquisitions. The thesis objective is to understand and analyse the current energy efficiency state and consequently propose some resource efficiency measures in order to replicate this new format for the new acquisitions to have a lower carbon footprint compared to the existing supermarkets belonging to the same brand.

The thesis project is an energy audit comprising analysis of historical data, site visit to current Azeri Retail's stores and consequent proposal of energy saving measures to be applied to the new acquisitions. In order to do this, three case studies have been developed: ATL, Project and REM.

The first step has been the literature review about the energy utilization and carbon footprint of the retails sector, past energy audits, current and future policy framework in Azerbaijan. During this phase, it turned out that Azerbaijan doesn't have a dedicated law for energy efficiency and therefore the construction phase doesn't consider energy efficiency a primary target. Furthermore, in average the refrigerators represent the 40% of the total energy consumption of the supermarket,

The site visit in Baku highlighted huge differences between the two brands' buildings owned by Azeri Retail: Fresco brand belongs to high structural and technological level supermarkets, while Sebet doesn't have any insulation on the envelope and the technical equipment is rather old.

Microsoft Excel have been used in order to create the tool to perform the calculations. The thermal losses considered are the transmission losses, due to the building envelope, and the ventilation losses, due to the temperature difference between the exhaust indoor air and outdoor air. The internal gains are included in the model, in particular occupancy and solar loads. Furthermore, every case study considers different technical equipment according to what it is representing.

All three built case studies refer to the geometry of the Fresco 2 building, since the comparison is more valuable if the buildings considered have similar geometry and activity inside. The ultimate aim of the modelling phase is to achieve good savings through measures in the REM case, since it is supposed to represent a useful list of saving measures to carry out during the actual refurbishment of the new stores. The measures proposed are: refurbishment of the envelope, purchase of a heat recovery heat exchanger, the exploitation of daylight through solar tubes and installation of LED bulbs, the refurbishment of cold rooms and the installation of double air curtains in the open refrigerated display cabinets. Great and positive results have been achieved during the modelling phase:

- Fresco stores can show very good structural properties and technological equipment and for this reason, each saving measures is additional to the already existing Fresco's buildings status.
- Although Fresco stores' buildings are better than Sebet stores' ones and better than average practice in Azerbaijan, huge savings have been identified and this means that

the new acquisitions could perform even better than the existing ones, mostly considering medium-low cost measures.

The report closes with the overall comparison between the three case studies' energy consumption and international benchmarks about food driven retails.

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I would like to thank Marco Morando, the Energy and Resource Efficiency Manager in RINA Consulting, that assisted me during the development of the project, helping me to strengthen my knowledge in the energy utilization field. He allowed me to work closely with the client, to learn technical procedures and managerial skills, making this thesis an amazing experience. I would like also to thank Cristina Bianchi, Technical Assistance & Sustainable Development Head, and Danilo Bosia, Resource Efficiency & Management Head, that introduced me in RINA Consulting, making this experience possible.
I would also give thanks to Hatef Larijani Madani, Professor at Royal Institute of Technology (KTH), who accepted to supervise me and he has been always present during the entire six months' thesis period. He gave me a lot of useful suggestions about how to structure and improve step by step my thesis work.

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1 INTRODUCTION

1.1 Background of retail buildings: energy utilization and GHGs emissions

The building sector is one of the energy sectors more discussed during the last decade. It consumed 30% of total final energy use and about 28% of direct and indirect CO2 emissions. The building construction sector, including the manufacturing of materials for building such as steel and cement, accounted for 6% of the global energy use and nearly 11% of the global CO2 emissions (1) . Summing up the two shares, the overall building sector is responsible for 36% of the final energy use and 40% of the total CO2 emissions in 2015. As reported in Figure 1, within this sector, the non-residential buildings (mainly commercial ones), accounted for 8% of the energy consumption (2) .

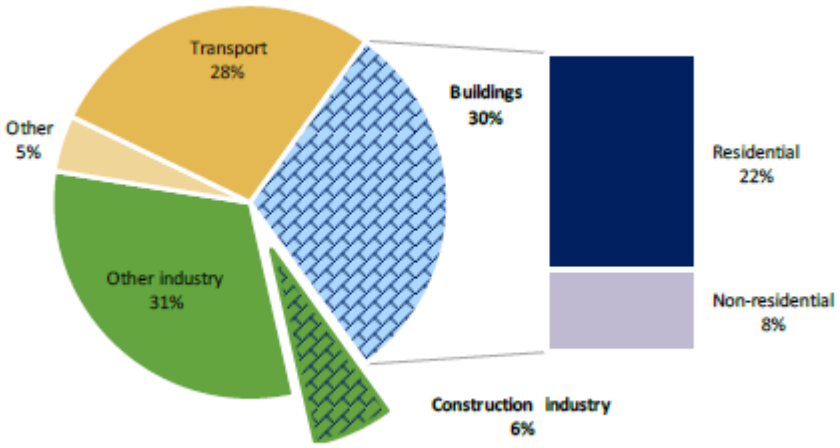


Figure 1 Share of global final energy consumption by sector, 2015 (2)

The commercial buildings sector comprehends a wide range of different types of buildings: retail stores, shopping malls, offices, hospitals, hotels, restaurants, sport facilities and others. The retail sector is interesting to study from an energy saving perspective. Just to mention, in 2010 it was responsible for the 47% of the total emissions from food chain refrigeration in UK (3) .

The thesis will be focused on an energy audit and consequent proposal of measures aimed for improving the energy efficiency in the retail buildings in Baku, Azerbaijan.

Supermarkets are complex energy systems in which several components interact, resulting in a large energy-user system. For instance, both in USA and France supermarkets have consumed 4% of the national energy used, in 2017. In Sweden this value accounts for 3% (4) .

The main equipment in this type of retail buildings is:

- Refrigeration system
- HVAC system (Heating Ventilation and Air Conditioning)
- Lighting

The refrigeration system is able to provide storage and food preservation to the products in a supermarket. Typically, depending on the type of products, this system works on two different temperature levels: low (around -20 °C) and medium (around 4 °C) temperature. The first one is used for frozen products, the second one for preserve chilled food.
The two main types of refrigeration systems are:

- Vapour compression cycle: it works with the aid of a compressor, driven by an electric motor, that sets two different pressure levels in the system. The other main components are the evaporator, in which the refrigerant fluid absorbs heat from the environment, the condenser, in which the refrigerant releases heat to the environment, and an expansion valve able to decrease the pressure of the refrigerant up to the evaporating pressure. In Figure 2 this thermodynamic cycle is represented.

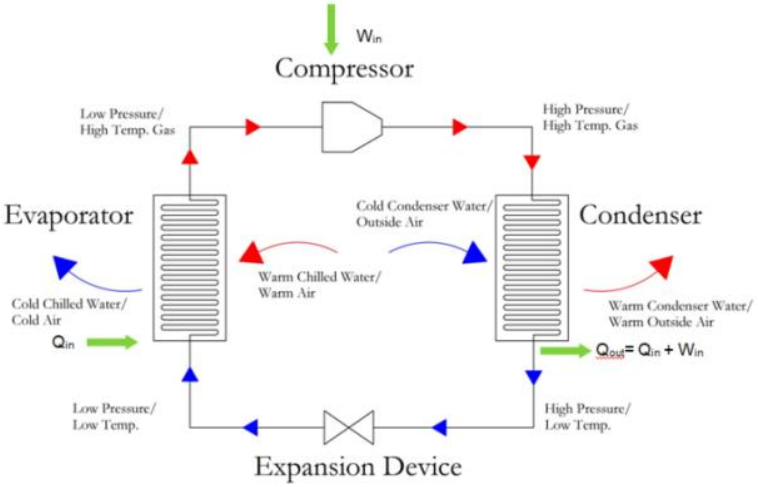


Figure 2 Basic vapour compression cycle layout

- Vapour absorption cycle: differing from the previous one, it works without the use of any compressor. Indeed, this thermodynamic cycle changes the gas back into liquid using only heat, no moving parts are involved in the process. The compressor is replaced by an absorber and a regenerator. Another main difference is that the refrigerant is dissolved in a suitable liquid, typically water. In Figure 3 the layout is presented.

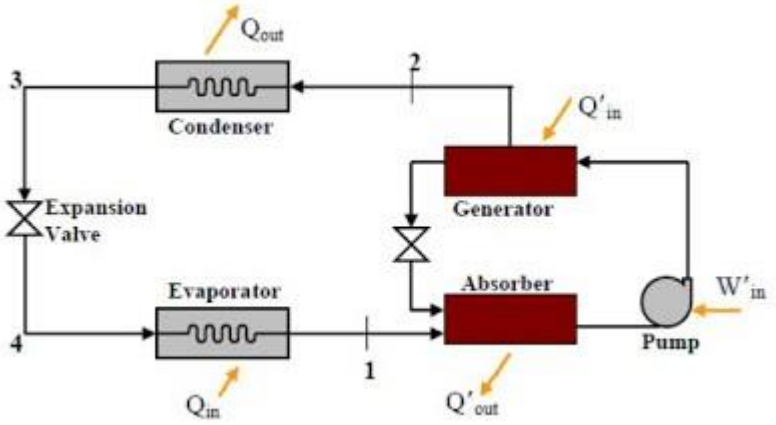


Figure 3 Vapour absorption cycle layout

The compression cycle is more common because of its higher coefficient of performance (COP). The absorption cycle is typically used when heat is available as a cheap source and electricity is unreliable, costly and unavailable.

The refrigeration system works with a fluid, called refrigerant, able to absorbs a huge amount of heat at low pressures during the phase change, creating refrigeration effect. The entire refrigeration system is crucial in studying the potential savings that a supermarket would be able to achieve. Indeed, approximately half of the energy consumption in supermarkets is associated with the refrigeration system (5). Refrigeration system has remarkable negative environmental impact due to greenhouse gases (GHGs) emissions: indirect emissions from electricity consumption and direct emissions due to leakages and refrigerant type. Although it is outside the thesis scope, several studies have been performed to estimate the environmental impact of different system configuration. (6) (7)

HVAC system has to provide a good indoor air quality, indoor thermal comfort to the staff and costumers in the supermarket. It is easy to figure out, that the well-being of these people can indirectly affect the sales of the stores, that's the reason why they are crucial in this analysis.

Lighting will be investigated as well. Being a component always working in the typical operating hours of a supermarket, is reasonable to think that there is margin for energy savings.

Since the demand in the overall building sector is increasing each year, the renewable energy production and the energy efficiency are being investigated in order to reduce the building's energy consumptions and related environmental impact.

1.2 Research questions & thesis objective

The thesis has been developed as a part of a larger project financed by the European Bank for Reconstruction and Development (EBRD). EBRD intends to finance Azeri Retail for the refurbishment of its new acquisitions in Baku. The brand already owns 13 stores divided in 7 "high quality supermarkets" (Fresco stores) and 6 "discount markets" (Sebet stores). The

project consists in the development of a new Fresco supermarket-format store for its six new acquisitions. These new stores are obtained, as stated before, by a refurbishment of existing building, which will be rented on a medium-long term by the supermarket brand.

Azeri Retail considers the current Fresco format store a good technological and structural level, therefore Azeri Retail would like to replicate this state also for the new acquisitions. The thesis objective is to understand and analyse the current energy efficiency state and consequently propose some resource efficiency measures in order to replicate this new format for the new acquisitions to have a lower carbon footprint compared to the existing supermarkets belonging to the same brand.

In order to assess the energy performance of the existing retails, an energy audit in Baku has been arranged. During this site visit, the audit team will gather as much information as possible in order to work on solid data and to build reliable case studies. The data will be about current materials used for the envelope of the buildings, current energy bills, energy technology available in the stores, with special focus on chillers, cold storage room and ventilation units. From this information, an energy model in *Microsoft Excel* will be created able to take into account the physical structure of the building, any installation, equipment and occupancy schedules and any item interesting for such analysis. Furthermore, the model will be used to highlight the shares of energy consumption by fuel and by energy source. The following step will be to propose REMs in order to investigate if there are margins for significant energy savings. REMs can be about both sustainable power production and energy utilisation. This means that both the technology and its operating parameters will be investigated. The analysis about measures' impact will be defined also through the typical economic procedures, such as NPV methodology, simple payback period and discounted payback period.

The three research questions addressed in the thesis work are listed below:

- *What is the current energy efficiency state of supermarkets in Azerbaijan?*
- *What are the most competitive efficiency measures for supermarket buildings in Azerbaijan?*
- *How can the Azeri Retail supermarkets be (partially) decarbonized?*

1.3 Thesis scope

The thesis investigates potential energy savings in the retails sector in Azerbaijan. As described in the above sections, this sector along with the overall building sector is responsible for a large part of GHGs emissions and energy utilisation. In particular, the supermarket sector has a large use of refrigerants, that are one of the main responsible substances for the ozone depletion layer and global warming.

Markets, supermarkets and convenience stores are intensive user of energy in all countries. Their energy consumption is estimated to be 2% in Poland and 4% in USA of total national

electricity consumption. The average energy consumption for supermarkets is 420 kWh/m^2 (Sweden), 570 kWh/m^2 (France) and 800 kWh/m^2 (Poland). For small convenience stores (less than 300 m^2 of floor area) it reaches 470 kWh/m^2 up to 650 kWh/m^2. There is a great potential for improvement of energy systems in stores. It may involve refrigeration systems, lighting and HVAC. Energy saving technologies such as heat recovery, defrost control system, energy efficient lighting, high efficiency motors can reduce considerably energy consumption both in small and big stores (8).

2 METHODOLOGY

2.1 Literature review

As all research activities, the main target is to define correctly the questions that would have to be solved. It is really important to define correctly the topic of the study, its environment and boundaries. There are different ways for collecting information, especially in the field of energy efficiency for existing buildings: from the authorities and government, from observations on sites, interviews and surveys.

For this master thesis, past energy audits in the supermarket sector have been reviewed. However, since most of them weren't performed in Azerbaijan or in a situation with similar energy system's characteristics, a lot of them have been scrapped since they didn't allow a direct and reasonable comparison. Nevertheless, interesting benchmarks about final energy consumption and breakdown of this have been collected in order to compare the results obtained in this project with other interesting studies. Furthermore, general information about the retail sector globally, in Azerbaijan, in Europe and in CIS countries have been gathered. From the website *Science Direct* a lot of studies about measuring and modelling energy use in an operational supermarket have been read and taken in order to have an idea of how this commercial sector works. As it is highlighted in this report, this commercial sector is different from the other buildings sector and therefore a proper literature review has been needed. Furthermore, historical and current status of energy policy framework has been reviewed to understand the historical and current willingness of Azerbaijan to take measure to improve energy efficiency in the country (9) (10).

It is important to highlight that in-house information from the project team of RINA Consulting have been used to be compared to data gathered along the literature review process. All this have allowed a further comparison between Azerbaijan and near Countries.

To conclude, since the core task of the thesis is to provide a list of resource efficiency measures for the future supermarkets of Azeri Retails, some key technologies and structural materials have been reviewed, according to the current state-of-art of the different technologies and energy needs of the shops, but also legal and policy framework currently present in Azerbaijan.

2.2 Data collected

In this section, the most important data gathered during the literature review and used for calculations and benchmarking will be reported. However, some data used in the model come from the energy audit performed, therefore won't be reported here but in the audit dedicated section.

The retail sector includes many types of stores. They are characterized by their average sales area and are normally classified as:

- Hypermarkets - 5000 m^2 to over 10,000 m^2 sales area;
- Superstores - 1400 m^2 to 5000 m^2;
- Supermarkets (mid-range stores) - 280 m^2 to 1400 m^2;
- Convenience stores less than 280 m^2.

The energy consumption of a retail outlet may vary considerably depending on the format and segment. Food retailing accounts for the highest specific consumption mainly due to the high power consumption for food refrigeration and merchandise presentation in the fresh products area, whereas in non-food retailing store lighting account for the biggest part of electricity consumption. In both types of retails heating, ventilation and air-conditioning systems account for about one third of energy used. (11)

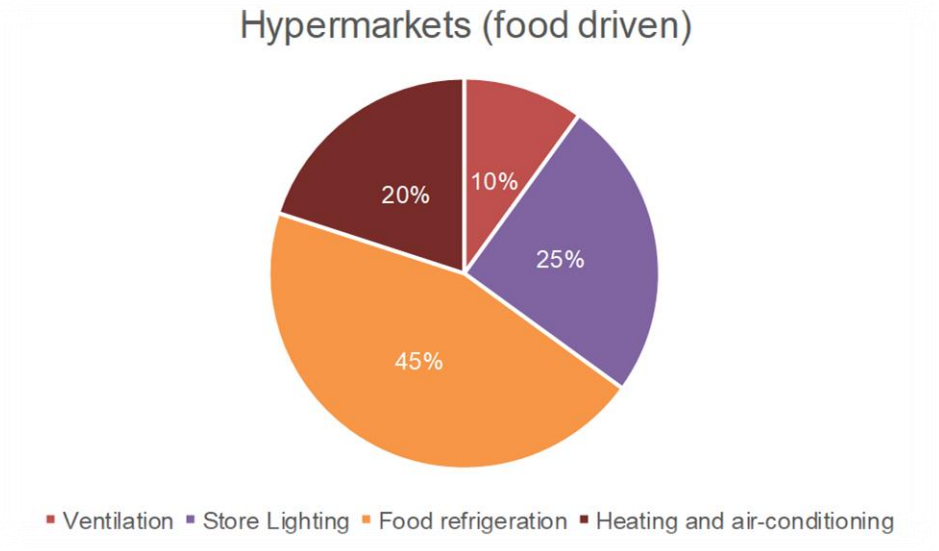


Figure 4 Electrical energy consumption in food retails

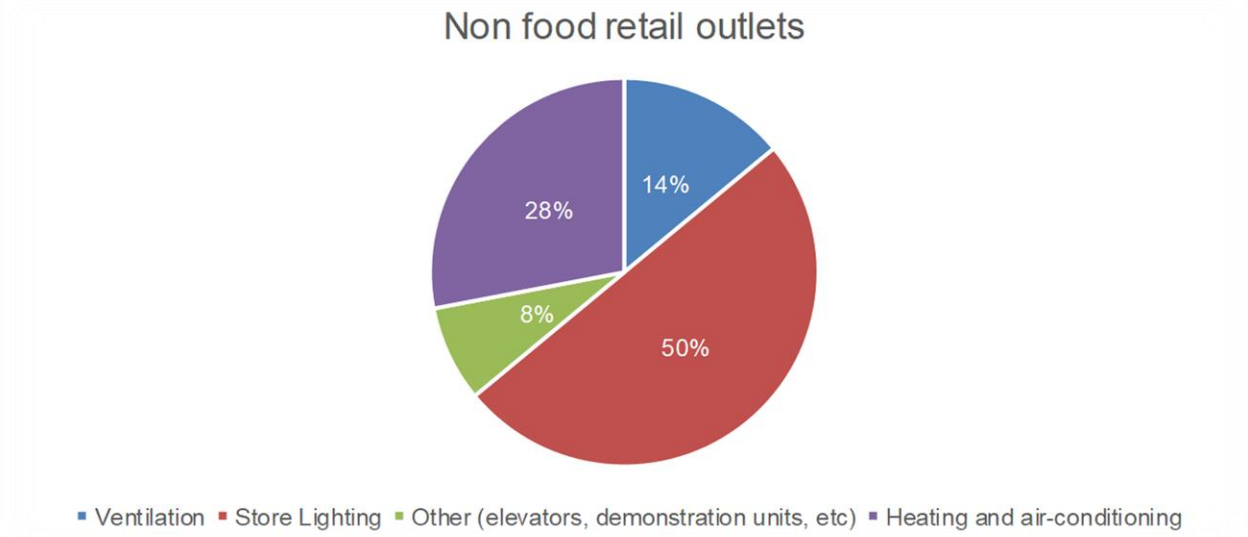


Figure 5 Electrical energy consumption in non-food retails

Interesting benchmarks about electricity and thermal energy consumption in supermarkets have been retrieved during the literature reviewed. These are reported in the below figures, expressed in kWh/m^2. They have been taken from International Sustainability Alliance (ISA)[1].

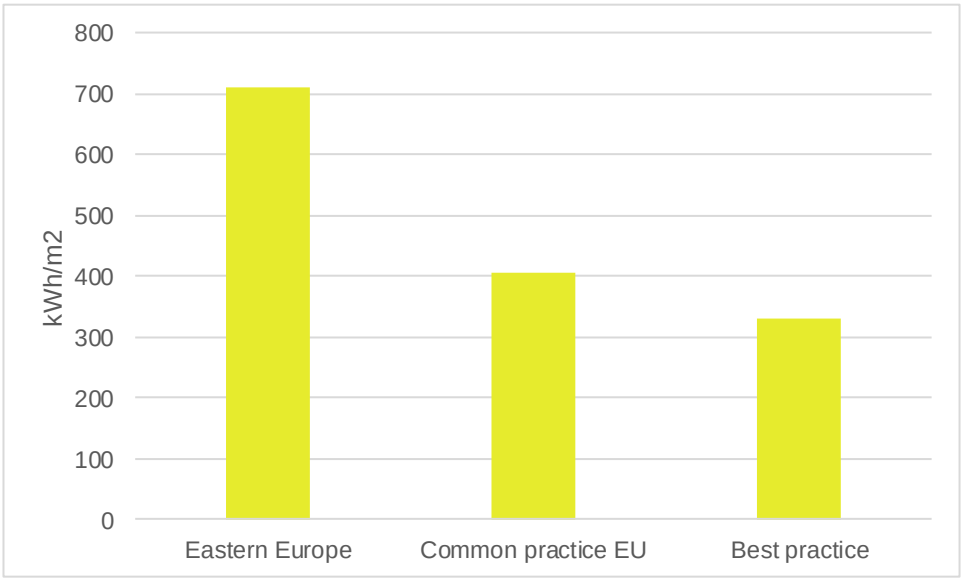


Figure 6 Electricity energy benchmarks (12)

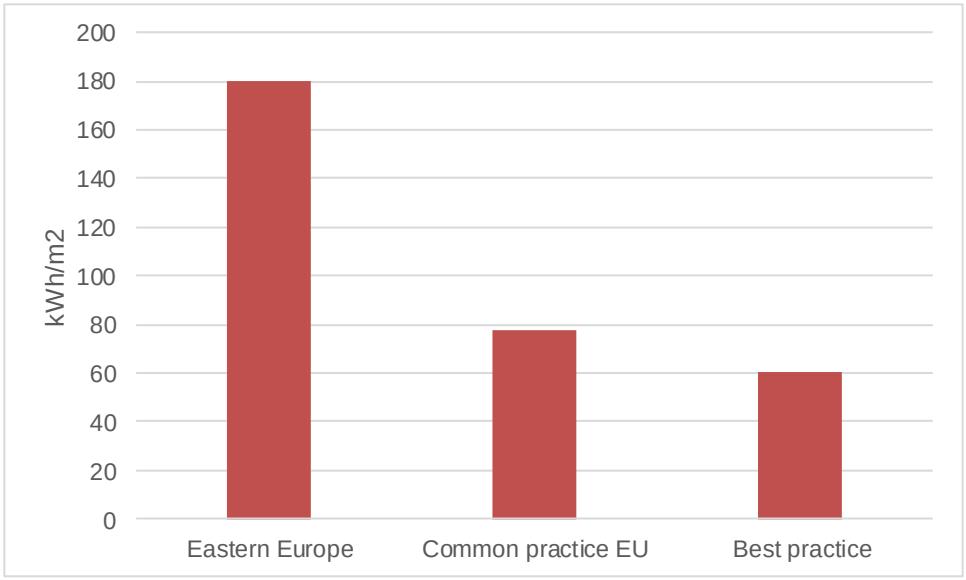


Figure 7 Thermal energy benchmarks (12)

In order to set the indoor conditions, conditions requirement in supermarket have been read and used. (12)

[1] In the graphs, Eastern Europe includes following countries: Bulgaria, Croatia, Czech Republic, Greece, Hungary, Kazakhstan, Moldova, Poland, Rumania, Serbia, Slovakia, Turkey, Russia and Ukraine:

Table 1 Conditions requirement in supermarket

	Customers	***Frozen food***	***Fresh fruit and vegetables***	***Cooled food***
Air temperature (°C)	19 - 21	-18	15	4
Relative humidity(%)	40-70	70	90	90

Although different types of systems and approaches have been tried to improve thermal comfort and reduce energy consumption, such as under floor heating, displacement ventilation and natural ventilation, the most common system nowadays is the all air constant volume system. This system has the advantage to supply air with low concentration of contaminants and to provide ventilation, heating and cooling by conditioning air in the central plant and distributing it through overhead ductwork to different parts of the store. Return air ducts return the air to the air handling units where part of it is mixed with fresh air and returned to the store whereas the rest is discharged to the atmosphere.

A last remark about windows: this is one of the key elements when estimating thermal losses and therefore thermal comfort in an indoor environment. During the literature review, some important data have been found, and therefore utilised during the analysis. (13)

Table 2 Types of glazing and related characteristics of windows

Type of glazing	***Filling gas***	***U-value (W/m^{2}K)***	***Solar factor g***
Single glazed	n.a.	5.9	89%
Double glazed	Air	2.6	75%
Double glazed low-e	Air	1.5	58%
Double glazed low-e	Argon	1.1	56%
Double glazed selective	Argon	1.1	31%
Triple glazed low-e	Argon	0.6	36%

2.3 Energy audit performed

There is not only one methodology of making a sustainable building. Retrofitting actions plan is a result of energy audits, which are never carried out in the same way. All the assessments and analyses are dependent of the objectivity of the engineers, and it has been admitted that there is still a difference in the real life between the proposal actions and the final retrofitting plan.

A core part of the project is the energy audit performed. As it will be explained later, the energy audit allowed to have a broad idea of the current situation of the two stores visited and considered for the final analysis. Energy bills, building structural elements, power installed, equipment used, general information about the weather and about the outdoor

temperatures have been gathered during the energy audit and during the survey carried out to the personnel of Fresco 2 store.
Energy audits are currently used as tools to asses an existing building. Thanks to them, it is possible to collect a large amount of information about the energy building consumptions, its systems (electrical, Heating Ventilation Air Conditioning, Information Technology, etc.) but also about its management and its built environment. There is not only one methodology for carrying out an energy audit, all the consulting society have a specific one, but they are based on the same elements, which are energy bills, walk through analyses and interviews of the managers of the site, measurements, building simulations, and benchmarking. Then, there is the step of developing a retrofitting plan. This one would be the base for highlighting the strengths and the energy potential of the building.

2.4 Model built

In order to develop a list of measures, an Excel model has been built for the purpose of this project. It is very complex and composed by several worksheets linked each other. The model is able to evaluate the heat balance for the three cases considered in the thesis and for the last one (REM case) the model estimate the benefits of the measures proposed and their cash flow diagram. It can be divided in three main parts: the first one containing very general information, the second one information about the building itself and the simulation of the first two cases and the third one with the simulation of the proposed measures. Most of the data inserted in the model have been gathered during the energy audit.
Some general inputs have been set such as grid emission factor of Azerbaijan, exchange rate from AZN to EUR, the primary energy factor for electricity and annual increase of price for electricity. Every value will be stated in the following sections.
After this set up, dimensions and materials of the building facades, windows, roofs, doors and floors have been listed because they are key parameter for transmission losses. Furthermore, desired indoor temperatures have been set for estimating the ventilation losses. The Baku's weather has been analyzing because of the estimation of the heating degree days. Concerning heat gains, equipment, solar radiation and occupancy have been considered for the estimation of heat balance. During the site visit a detailed survey has been carried out with the personnel of the supermarket in order to gather information about occupancy. Valuable information has been provided and therefore used in the model. To conclude about equipment, lighting and refrigerators models have been listed during the site visit and the consumptions have been estimated accordingly. In this section of the model, the two first cases (ATL case and Fresco 2 case) are simulated.
The third part consists in the application of the proposed measures in order to estimate the new consumption of the building. Energy efficiency and related cash flow analysis has been carried out.

In the below figure a very simple scheme of the block flow diagram thought for the project is represented.

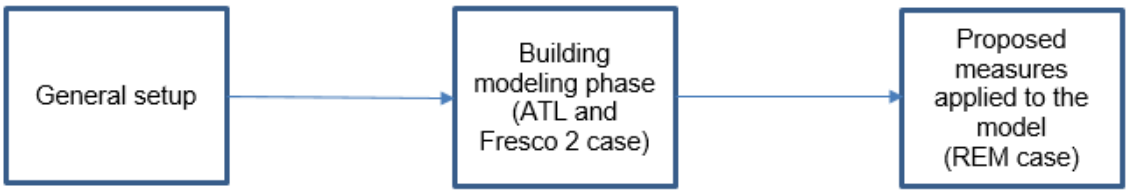


Figure 8 Simplified representation of the block flow diagram of the model built

The limitations and benefits of the model will be discussed during the discussion part of the report.

The output of the model will be compared each other for a first useful comparison. Furthermore, data gathered during the literature review will serve as benchmark and a second comparison will take place.

2.5 Case study analysis

The case study has been based on the developed model and methodology. Also, it would take into consideration the benchmarks and important points introduced through this thesis. Three cases have been defined: ATL case, Fresco 2 case and REM case, in order to compare the outputs of the model internally, between the three of them, and externally with benchmarks retrieved. Indeed, despite the great potential of energy audits, a benchmarking process is essential. This step is a great mean of understanding correctly, what are the weaknesses and advantages of a selected building sector.

3 AZERI RETAIL SUPERMARKETS

3.1 General overview of the company

Azeri Retail LLC is a food supermarket chain operating in Baku, Azerbaijan. It currently owns 13 stores under two separate brands Sebet and Fresco. One of the main differences between these two brands is the quality of the food present in the store. In Sebet stores, cheaper food is present, while in Fresco ones is possible to purchase higher quality products. There are 7 Fresco stores and 6 Sebet stores.

In Table 3 is possible to see the list of the different stores, the opening date, their gross and sales area.

Table 3 Azeri retail 13 existing stores

Name of the brand	***Opening date***	***Gross area (m^2)***	***Sales area (m^2)***
Fresco 1	09/08/2012	1,166	754
Fresco 2	12/02/2013	2,150	1,660
Fresco 3	16/09/2013	1,321	1,037
Fresco 4	10/04/2014	1,300	907
Fresco 5	21/10/2014	1,627	1,159
Fresco 6	20/02/2015	2,357	1,700
Fresco 7	29/04/2015	1,373	957
Sebet 1	27/05/2011	540	377
Sebet 2	20/03/2011	695	454
Sebet 3	22/04/2011	602	438
Sebet 4	10/05/2011	514	375
Sebet 5	28/05/2011	494	355
Sebet 6	19/03/2012	727	556

As can be seen in Table 3 all existing stores were recently opened (the oldest was opened 8 years ago). During the audit it has been possible to figure out that technologies found in Fresco and Sebet stores differ a lot. In the first ones, new and high performance technologies were present, while in Sebet stores rather outdated equipment was installed. This difference reflects the choice of Azeri Retail to have cheap supermarkets (Sebet) and higher level supermarkets (Fresco). This choice has implied larger energy consumption but smaller installation costs for Sebet, vice versa for Fresco.

Fresco Retail stores primarily sell food, including fresh products selling as fishery, butchers, delis, and bakeries. In addition, they provide non-food products, mainly household items.

4 ENERGY AUDIT AND MAIN FINDINGS

4.1 Different types of energy audit

The energy audit is a systematic procedure to obtain an adequate knowledge of the profiles of energy consumption of an existing building or group of buildings, an industrial and service private or public, in order to identify and quantify in terms of cost effectiveness of energy saving opportunities and the relationship of what is revealed (14). In order to conduct a proper energy audit, specific standards need to be followed. One of the most common standards are the ASHRAE standards.

Following these standards, it is possible to identify four different energy audit levels:

- Preliminary Energy-Use Analysis (Benchmarking)
- Level 1: Walk-through
- Level 2: Energy Survey and Analysis
- Level 3: Detailed Survey and Analysis

The preliminary energy-use analysis mainly includes an analysis of the utility bills, summary of utility bill data, calculation of energy use intensity (EUI) and a benchmarking to compare the site's energy usage to that of similar sites in the same region, if possible.

The Level 1 energy audit is walk-through audit that will identify energy efficiency measures (EEMs) while limiting the necessary engineering time and costs to produce the report. The brief report focuses on low-cost and no-cost measures, although capital measures are identified when found.

The Level 2 energy audit is more detailed and requires proficiency and thought to create a quality audit report. It includes a complete description of the facility, including an equipment inventory, an energy balance, detailed energy savings and costs associated with each low-cost and not-cost measure, financial analysis of each recommended measure, identification and rough estimates of capital project costs and savings.

The Level 3 audit is designed to provide additional engineering rigor for more expensive capital projects where risk is less tolerated. For instance, calculations for HVAC measures are done with hourly simulations. Typically, a scope of work and schematics are provided with the audit so that the contractors installing the measures understand exactly what is to be installed (15) .

Following the ASHRAE standards' definitions, in the thesis project a Level 2 energy audit will be performed.

4.2 Main findings from Fresco and Sebet retails

In agreement with Azeri Retail, the audit team visited 3 stores: two Fresco stores (Fresco 2 and Fresco 5) and one Sebet store (Sebet 1). During the audit, what was anticipated from Azeri Retails has been confirmed: Fresco and Sebet stores are at two different technological levels. It was interesting to note that Fresco 2 and Fresco 5 have equipment belonging to similar technological level and this will be extremely useful for developing the case studies. From now on, Fresco 2 will be the reference for the Fresco stores, and the Sebet 1 for the Sebet ones.

4.2.1 Sebet 1 store

The Sebet 1 store has a gross area of 540 m^2, out of that 377 m^2 is sales area. The store has been opened in May 2011 and it works everyday between 08:00 and 23:00. It is located in the ground floor of a multi-story building, having a limited window area. Windows are also used to advertise objects available in the supermarket and this causes a further reduction of daylight in the store. The glazing is single pane. Regarding insulation, walls and roof do not include any specific kind of thermal insulation.

Inside the store, six LG air conditioner were present for heating and cooling. No supply ventilation system is present. The lighting fixtures in the sales area are tubular fluorescent lamps (36 W and 58 W). Unfortunately, no further technical information about the equipment has been given during the site visit.

During the audit, the monthly electricity consumption in 2018 has been given. It is reported in Table 4.

Table 4 Sebet visited store monthly energy consumption

Month	***Electricity Consumption (kWh/m2)***
January	39.8
February	27.7
March	25.0
April	56.7
May	28.6
June	50.4
July	40.0
August	47.1
September	48.5
October	36.2
November	45.3
December	37.3

The customer occupancy in average is 6,741 per week, this means 1.8 customer per square meter per day.

4.2.2 Fresco 2 store

The Fresco 2 store, opened in February 2013, is located at Nizami 129 in Baku. The climate condition of Baku can be classified as moderately humid subtropical. Baku has a subtropical semi-arid climate with hot and dry summers, cold and occasionally wet winters, and strong winds all year long. The daily mean temperature in July and August averages 26.4 °C and there is very little rainfall during that season. Winter is cold and occasionally wet, with the daily mean temperature in January and February averaging 4.3 °C. Winter snow storms do occur but are rare and snow usually remains only for a few days after each snowfall.

The total area of the building is around 2,150 m^2, which 1,660 m^2 for sales and around 490 m^2 for the storage and cold rooms. The height of ground floor is equal to 3.5 m and the total estimated net heated volume is about 6,400 m^3. The store is located at the ground floor of a multi-story building and it is open every day between 08:00 and 23:00. Concerning the envelope, three sides of the store consist of glazing. The entrance faces north.

Although it is not one of the newest stores owned by the Azeri Retails, the store is fitted with the best technology available to the company and provides a high level of comfort to its customers. During the site visit, specific information about the building envelope, HVAC system and lighting has been provided.

Building envelope

The building is constituted by a reinforced concrete structure and the main facade of the building is made by double glazing windows (thickness 20 cm, air 100 %) and their U-value is equal to 2.6 W/m^2K (13). As in the Sebet store, the external walls do not present any kind of thermal insulation. The material is plastered bricks and reinforced concrete (thickness of about 50 cm). Therefore, the U-value is estimated to be 0.8 W/m^2K. The roof is made of reinforced concrete, without thermal insulation, and 1.5 mm of roofing membranes. Therefore, the estimated U-value for the ceiling is 1.2 W/m^2K. The ground cover at the basement floor is composed by reinforced concrete, cement finish and ceramic tiles, without thermal insulation. Therefore, the estimated U-value for the ground floor is 1.3 W/m^2K. The main entrance facing north is equipped with automatic door, occupying an area of 6 m^2. Its U-value is estimated to be 2.7 W/m^2K.

The U-values of the building envelope don't reflect any energy efficiency measure adopted in the past. Indeed, given the abundance of gas and oil resources in Azerbaijan, energy efficiency in the construction sector does not seem to be a concern. However, some governmental bodies addressed this issue within the frameworks under their competence. For example, the Ministry of Environment has the responsibility of identifying new sources of energy, and the Ministry of Energy is responsible for the energy supply of all buildings in the country.

In 2004, the State Programme on the Use of Alternative and Renewable Energy Sources in the Azerbaijan Republic (16) for 2005–2015 was adopted by a Presidential decree. This programme concentrates on developing new energy resources; however, it covers the construction sector only to a very limited extend. Action 15 of this programme urges different ministries to cooperate in providing buildings with alternative energy resources, prompting them to provide heat to residential, institutional, and other buildings by using the thermal energy of geothermal waters (2005–2013). The Ministry of Energy is also working to forecast heating and energy demands relating to ongoing construction projects, as it manages the country's gas and oil reserves. During the last 5–6 years, meters for energy and gas consumption have been installed almost everywhere, which has led to a minimization of loss and a more rational use of these services by the population.

There is no separate law on energy efficiency, but there is a Law on the Utilization of Natural Resources (1996). Some amendments are planned for this law, which also covers the issue of advanced technologies for producing and transporting energy. The Law also addresses the development of alternative energy resources. A very important activity in terms of saving energy in buildings is the mass replacement of existing windows by more efficient ones. In addition, architectural planning has an important role in broader efforts to enhance energy savings, through better insulation and correct solar orientation.

HVAC system

The store is equipped with a rooftop unit and two variable refrigerant flow (VRF) units that provide heating, cooling and ventilation at the same time. The ventilation system is connected to a McQuay rooftop unit, with a flow rate varying from 14,100 – 30,000 m^3/h.

The VRF McQuay is fed by R410a refrigerant. The flow rate varies from 11,600 – 22,000 m^3/h.

Inside the store there are 26 cassette type fan coil units that provide the main heating and cooling effect. There are two different types of units installed.

It is interesting to highlight that these units are manually controlled, using a remote control that the shop personnel can use.

To conclude with the HVAC system, the hot water service (HWS) is provided by a small centralised electrical boiler. Unfortunately, because of the position was not possible to read its size during the site visit. Since the water cubic meter consumed per year has been given during the site visit (approximately equal to 830.7 m3/year), knowing the hours of operation of the boiler, it has been possible to estimate the annual consumption of the boiler.

Lighting

The lighting of aisles consists of high efficiency TL5 Philips fluorescent lighting. These fixtures are supported by spotlights in special points of interest in the supermarket, as the bakery counters. The overall electrical capacity of the lighting system is about 25 kW for

indoor area. The calculation will be further explained in the lighting energy consumption calculation.

Refrigerators, freezers and cold storage room

The major energy service load is constituted by refrigerators, freezers and cold storage rooms. There are two cold storage rooms: one kept at a +2, +4 °C, and another kept at -22 °C. Both the cold storage rooms are insulated by an insulated panel surface with a thickness of 8 cm. The cold rooms and the refrigeration cabinets inside the store (with the exception of four low temperature fridges) are all connected to one central chiller located outside the storage area. The low temperature fridges are closed cabinet type. Furthermore, the glass doors are equipped with an electric heater. There are also cabinets for drinks: these are fitted with triple glazing doors to reduce heat gains.

All other open refrigerated display cabinets (ORDC) fitted with insulation curtains that are deployed manually only at night but are not equipped with air curtains, and therefore remain opened for the entire opening hours. A summary of the equipment related to the refrigeration activity has been gathered and used in the modelling phase..

Starting from this amount of information about the two stores, it has been possible to develop three case studies. Their definition is explained in the following chapter.

5 CASE STUDIES

The main task of the thesis project is to propose resource and energy saving measures to Azeri Retail for the new supermarket acquisitions. Three case studies have been developed:

- ATL **Errore. L'origine riferimento non è stata trovata.**case: this case study represents a good reference of the common technological level for retails in Azerbaijan.
- Fresco 2 case: it basically represents the current condition of the Fresco 2 store. It is supposed to present better performance than the ATL case, since during the audit, Fresco 2 store seemed to represent the high level retail standard from the technological point of view.
- REM case: this case is a further improvement of the Fresco 2 state. Here, several resource efficiency measures are proposed in order to explore the potential savings that the new supermarket could be able to achieve. As it will be explained later, each measure propose has a starting point the current Fresco 2 state.

In general, the performance of each case has been assessed considering the building's space heating, cooling, ventilation, lighting, domestic hot water and refrigerators, taking into account losses through the external envelope and ventilation, the internal and solar gains and climatic conditions. In the report, thermal losses, thermal needs and energy consumption are calculated. In details, thermal losses are the heat losses related to the structure of the building and to the temperature difference between inside and outside. The thermal need is the sum between losses and gains, considering also heat recovery equipment installed, if any. The energy consumption is how much energy is needed in order to satisfy the thermal needs of the building. The seasonal performance factor (SPF) is a measure of the operating performance of an electric heat pump heating (or cooling) system over a year. It is the ratio of the heat (or cooling) delivered to the total electrical energy supplied over the year. Obviously, the higher the SPF value the more energy efficient the system is.

$$Q_{th.need} = \left(Q_{loss,transmission} + Q_{loss,ventilation} \cdot (1-\eta)\right) + Q_{gains}$$

$$SPF = \frac{Total\ heat\ (or\ cooling)\ output\ per\ annum\ [kWh]}{Total\ electricity\ per\ annum\ [kWh]}$$

where clearly losses and gains have opposite sign.

Furthermore, in order to give more value to the proposal of a measure, the discounted net present value (NPV) has been evaluated. In finance, discounted cash flow (DCF) analysis is a method of valuing a project using the concepts of the "time value of money". All future cash flows (FV) are estimated and discounted by using cost of capital (r) to give their present values (PVs). The sum of all future cash flows, both incoming and outgoing, is the net present value.

$$NPV = \sum_{n=0}^{N} \frac{FV_n}{(1+r)^n}$$

5.1 ATL case study

The ATL is considered a good representative of the country ATL technological level for supermarket retail stores and it is used for understanding the current performances of Fresco 2 compared to the country's standards. Due to this choice, the geometry of the building is taken from Fresco 2 and the technical parameters are taken from common practice for retails in Azerbaijan, given by the RINA local office. The ATL is used across the report for comparison with Fresco 2, in order to highlight the improved energy and resource efficiency features of the company with respect to a reference minimum technological level in Azerbaijan.

5.1.1 Thermal losses

Transmission losses

Transmission losses depend on the insulating property of the element under investigation, as the envelope of the building.

In Table 5 there are the U-values used (again, taken from common practice for retails in Azerbaijan, except for windows).

Table 5 Common practice U-values for retails in Azerbaijan

Envelope component	***Common practice U-value (W/m^{2}K)***
Window	5.9
Wall	2.6
Roof	2.0
Floor	1.3

Using the area of the facades, the transmission losses can be evaluated as:

$$Q_{transmission} = U \cdot A \cdot HDD \cdot 24$$

Where $U \cdot A$ is the total transmittance of the building, and HDD are the "monthly heating degree days". The heating degree day is the difference between the desired average temperature inside the building and the average outdoor monthly temperature, multiplied by 30 since it is a monthly value. During the modelling phase, the desired temperature inside the store has been set as 21°C, both in winter and in summer.

Ventilation losses

Regarding the ventilation losses, common values are not applicable because the sizing of the ventilation system varies a lot according to retail's surface, products sold in it, choice of type of equipment used, occupancy in the stores, recovery heat exchangers and other characteristics. Because of this, equipment in Fresco 2 have been taken as reference.

The ACH requirement for the ATL supermarket is 5. Starting from this value, the design flow (44,760 m3/h) has been evaluated. In the ATL, no heat recovery is present.

To give a summary of the thermal losses calculated, the below Figure reports the monthly values and the total of transmission and ventilation losses.

The thermal losses are presented with the minus sign if they refer to an outdoor air temperature that is colder than the indoor air temperature, with the plus sign if the outdoor conditions are warmer than the indoor ones. This is valid throughout the report.

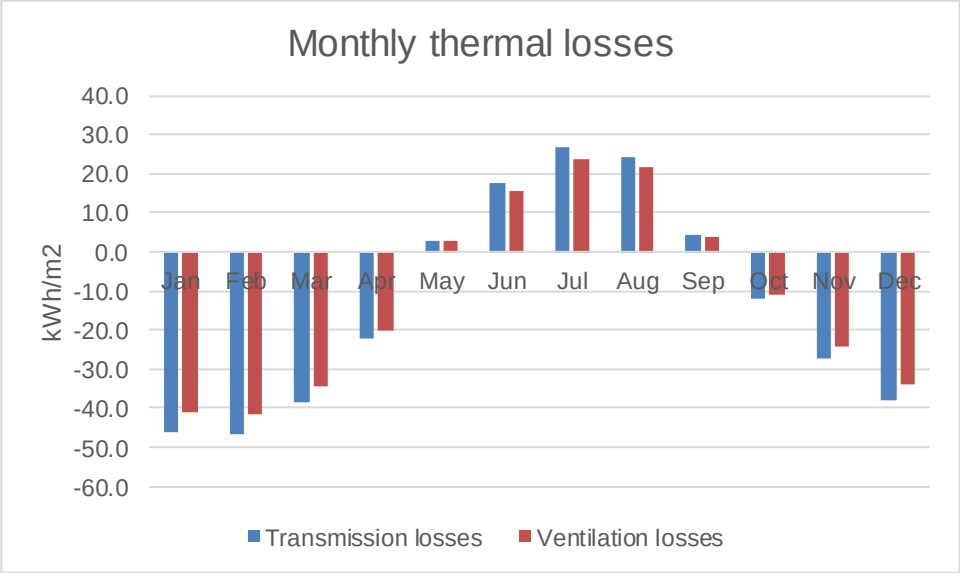


Figure 9 Monthly thermal losses ATL

5.1.2 Internal gains

After assessing the amount of losses in the building, the calculation of internal gains is the following step.

Person heat gain

The average heat gain for a person is around 70 W (assuming "seated or light work" value) (17). During the audit, Azeri Retail gave an average of client occupancy per week: 23,364 people/week. Every hour, the clients will be 223, obtained just dividing the occupancy per week by the hours of occupancy and by 7 days/week. Furthermore, in the supermarket there are also 20 people as staff. In total the heat gain due to occupancy will be

$$Q_{occupancy} = q_{person} \cdot n.people = 70 \cdot 243 = 16{,}976\ Wh = 17\ kWh$$

Solar heat gain

Since no energy modelling software has been used during the thesis, the daily radiation, in each direction, has been obtained using the website http://re.jrc.ec.europa.eu/pvgis/apps4/pvest.php#.

In the ATL, the radiation through windows has been set as 75% (18). This value comes from the reading of the scientific report of Ali Bahadori-Jahromi et al. in which specifies the value for the transmission of a clear glass type (without any selective films).

The monthly solar heat gain has been evaluated as:

$$Q_{solar} = R_{radiation\ through\ window} \sum (A_{glass,dir} \cdot Q_{rad,dir} \cdot (100 - S_{shadowing})$$

To conclude, the below Figure summarises the total internal gains present in the ATL, expressed as kWh/m^2.

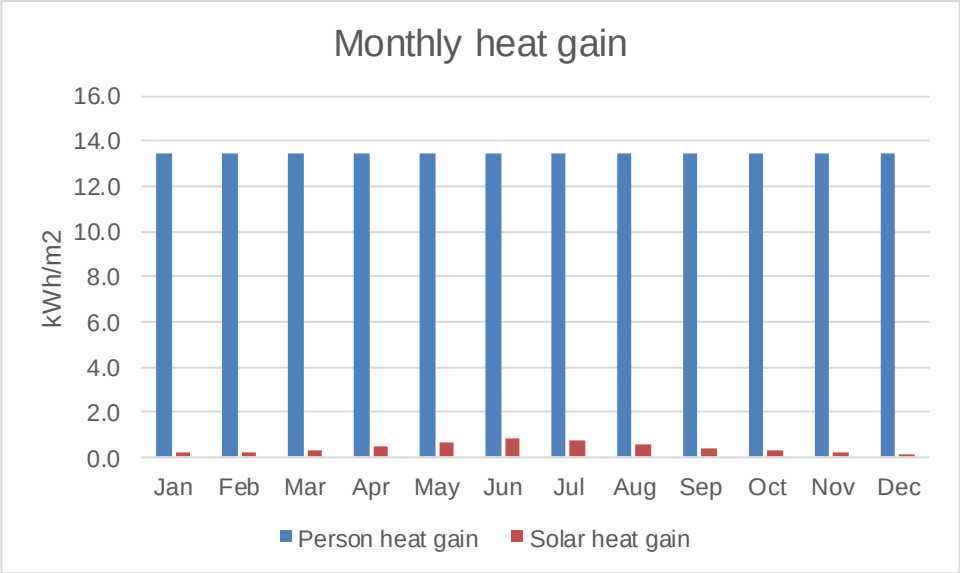


Figure 10 Monthly heat gains ATL

It is important to highlight that, being at the first floor of a multi-story building, and having a limited glazed area, the solar heat gain is extremely lower than the person heat gain.

5.1.3 Heat balance

The heat balance considers all the thermal losses and gains above mentioned.
Figure 11 plots the heat and cold losses during the year. In the chart, the cold losses include also the contribution for de-humidification for internally produced and fresh air moisture.

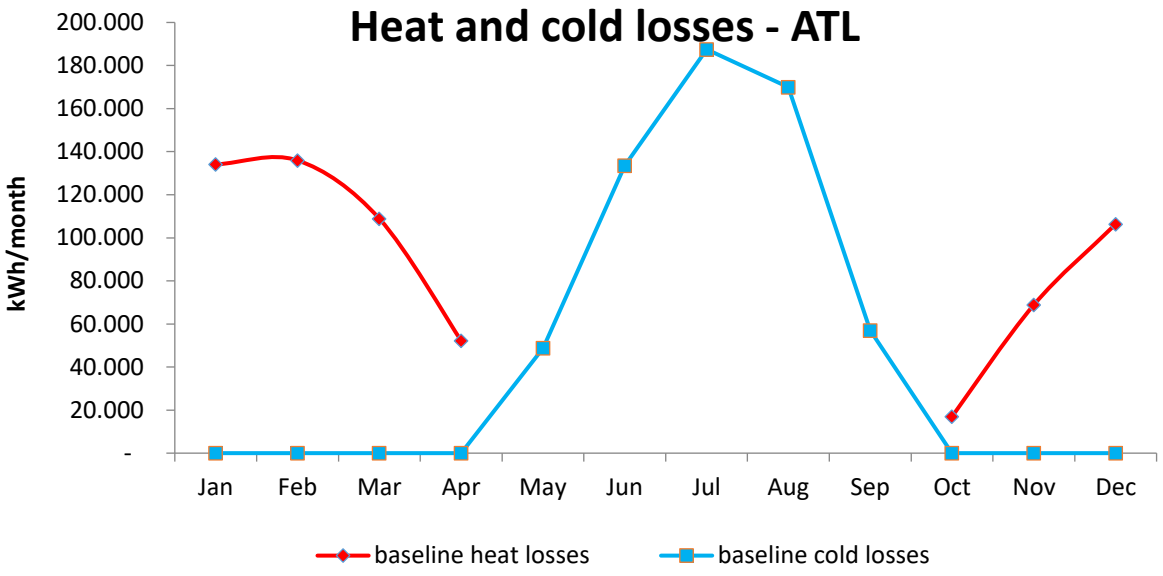


Figure 11 Energy balance for ATL

From this energy balance, the energy consumption for the ATL will be calculated.

5.1.4 ATL energy consumption

In this paragraph, the energy needs for matching the heat balance will be investigated.

At the beginning of this analysis specific parameters have been gathered in order to do a proper analysis: in general, for heating and cooling estimation the SPF of the equipment is considered, while for electricity consumption of the ventilation system the specific fan consumption is used, being the fan the most electricity consumer in a ventilation item.

The primary energy consumption and the end use consumption of the ATL is given in the below figures.

During the energy audit, valuable data have been gathered about lighting in Fresco 2. There are 584 TFL lamps, having an electrical power of 43 W. This means that in the Fresco 2 case the total lighting power will be 25.11 kW. Starting from this value, the necessary power in order for the ATL case to have the same illumination, or lumens, have been calculated. Indeed, the number of lamps installed doesn't change moving from one case to another. What changes is the type of lamp and the related efficacy, meaning how many lumens the lamp can generate using 1 Watt. For the ATL, being CFL the main lighting technology in Azerbaijan, 60 lm/W has been taken, for Fresco 2 100 lm/W (TFL lamps). (19)

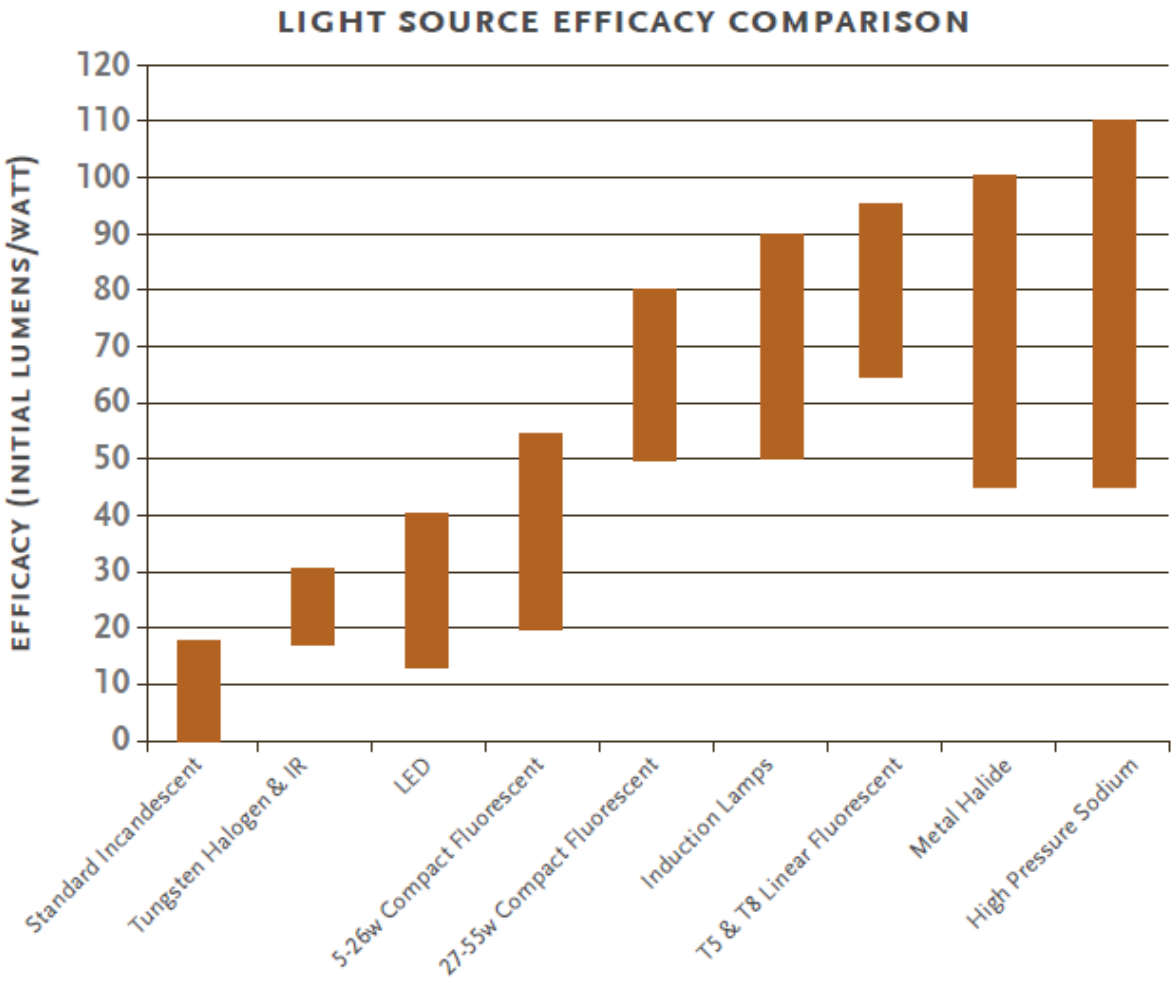


Figure 12 Light source efficacy comparison

The lumens production is 2,511,200 in both cases, but in the ATL case the power needed is 41.85 kW and in the Fresco 2 is 25.11 kW. Considering a contemporaneity factor of 90%, 15 h/day (since the store is opened from 8:00 – 23:00), meaning 105 h/week and 52 weeks, the total electricity consumption for lighting is 205,667 kWh/year, or 113 kWh/m^2 year.

The refrigerator found in Fresco 2 have lower annual demand than the average refrigerators used in retails in Azerbaijan. The performance and general information of such refrigerators is listed in a dedicated worksheet of the model. Considering utilization factors, air curtain present (240 W) and other assumptions regarding the load regulation of the refrigerators, the total energy consumption is 852,309 kWh/year, or 466 kWh/m^2 year.

To conclude the study of the energy consumption, the usage of hot water service has been evaluated. The yearly hot water consumption was given, 830.772 m^3/y. The hot water temperature, for sanitary, hygienic and thermal reasons (20) (21) (22), has been considered as 60°C and a cold one of 10°C:

$$Q_{HWS} = V \cdot c_{p,w} \cdot (t_{hot} - t_{cold}) = \frac{830.772 \cdot 4.19 \cdot (60 - 10)}{3.6} = 48{,}300 \frac{kWh}{year}$$

$$= 26 \frac{kWh}{m^2} year$$

The primary energy consumption for the ATL is presented. Since the overall energy consumption is matched through electricity consumption, the primary energy conversion was needed. In order to do this, a Primary Energy Factor (PEF) of of 2.43 $kWh_{primary}/kWh_{el}$ has been used. The national grid emission factor is 831 ton CO_2/GWh (23).

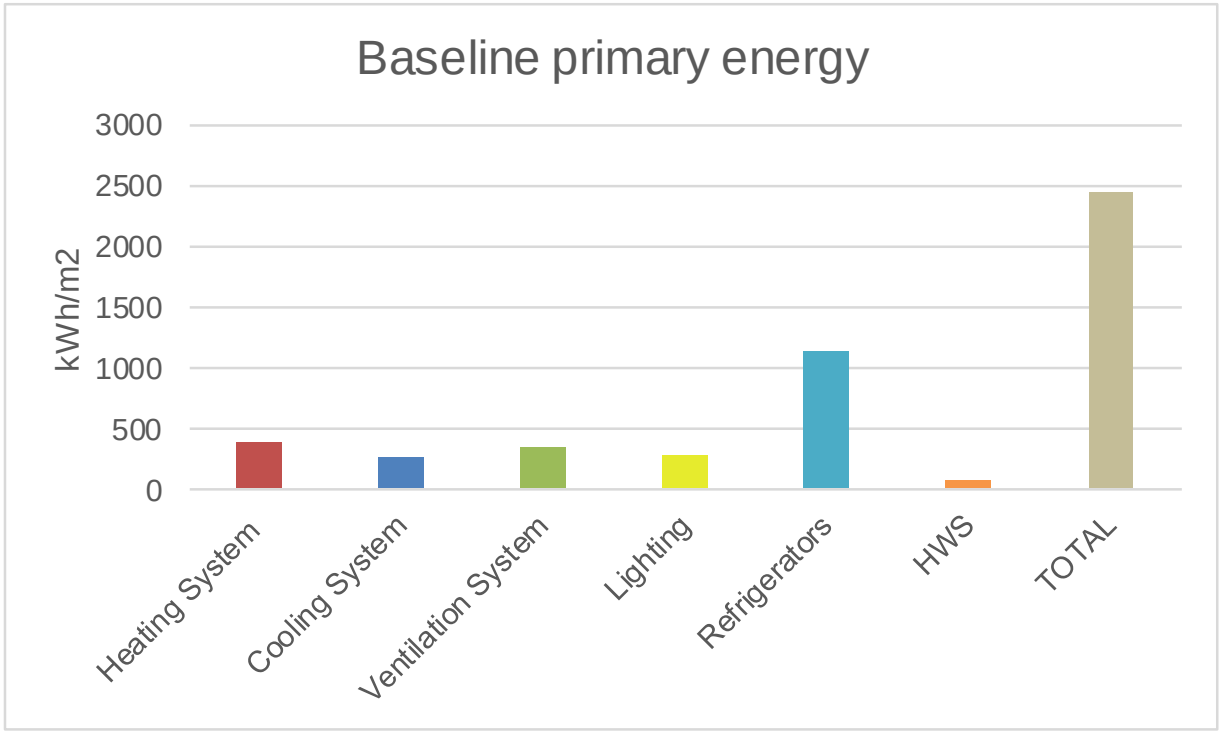


Figure 13 Primary energy consumption in ATL

To conclude, the end-use breakdown is presented in the below figure.

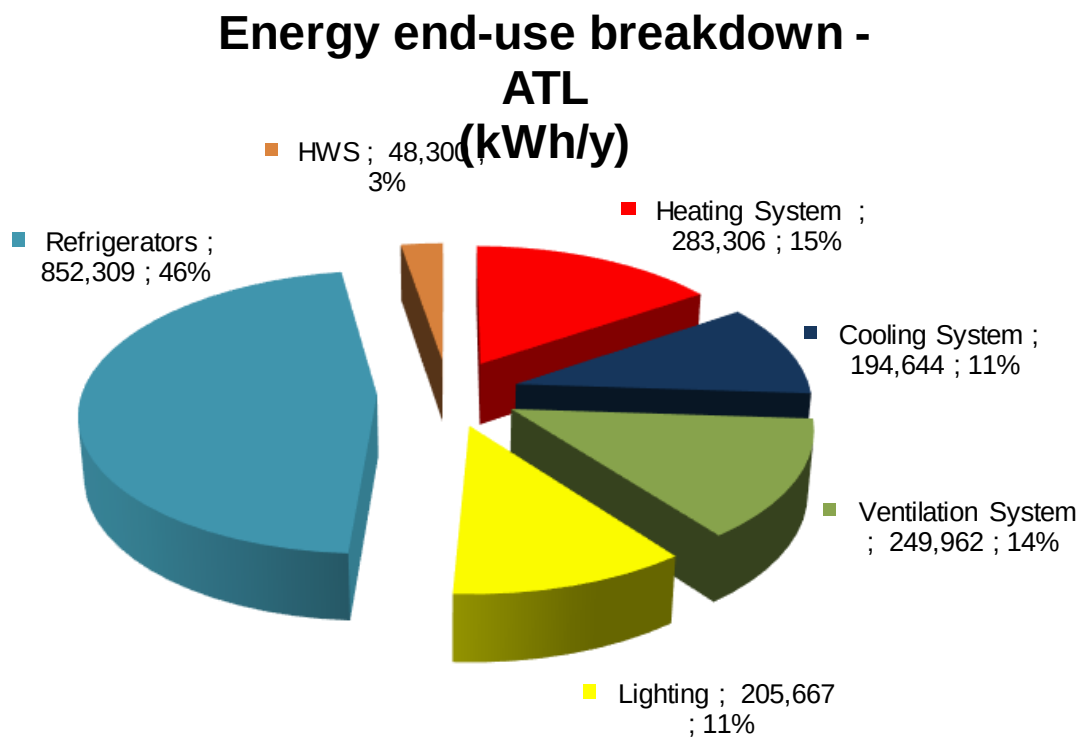


Figure 14 Energy end-use breakdown of the ATL case study

As the Figure 14 reports, the refrigerators' share is almost half of the overall energy consumption. This has been confirmed also by the literature reviewed (5) and reported in the related section. Furthermore, this confirms the robustness of the model built, and it allows to proceed with the study of the Fresco 2 case study.

Now it is time to compare this values with the benchmarks gathered. It is very important that the values presented and compared with the benchmarks are normalised, i.e. expressed in kWh/m^2. Expressing only the consumption as kWh it is impossible to understand if a supermarket (or in general a building) is more efficient than another.

The below table clearly states that current common practice in Azerbaijan is far less efficient that common practice in Europe or Eastern Europe countries. This gives more value and more sense to the following analysis carried out in the thesis project.

Table 6 Benchmarking of the ATL energy consumption value

Building	***Final energy use (kWh/m^2 year)***
International Benchmark (Canadian, USA) (24)	800
Eastern Europe Benchmark	890
Common practice in EU	483
Best practice	390
ATL	1,004

From the benchmarking analysis turned out that the common practice in Azerbaijan is a very poor and inefficient case. According to the benchmarks gathered, it consumes twice what is the consumption in Europe.
In the next section, a similar analysis will be performed for the current state of Fresco 2.

5.2 Fresco 2 case study

The Fresco 2 case study represents Fresco 2 store, as it is. All the structural and technical features are taken from data gathered during energy audit or given by technical staff of the company.

5.2.1 Thermal losses

Transmission losses

In Table 7 there are the U-values used in the Fresco 2 case. These data have been gathered and estimated during the energy audit in Fresco 2.

Table 7 U-values for the Fresco 2 case

Envelope component	***Fresco 2 U-value (W/m²K)***
Window	2.7
Wall	0.8
Roof	1.2
Floor	1.3

As can be noticed from the Table 7, the envelope of Fresco 2 is far better than the common practice in Azerbaijan. Indeed, in that case the total transmittance was 7,025 W/K, leading to higher losses, and in the Fresco 2 the building transmittance is 4,460 W/K. This is directly reflected in the below Figure.

The desired indoor temperature (21°C) haven't been changed with respect to ATL. This reflects the willingness of the Azeri Retail to offer a good thermal comfort to the customers in the store.

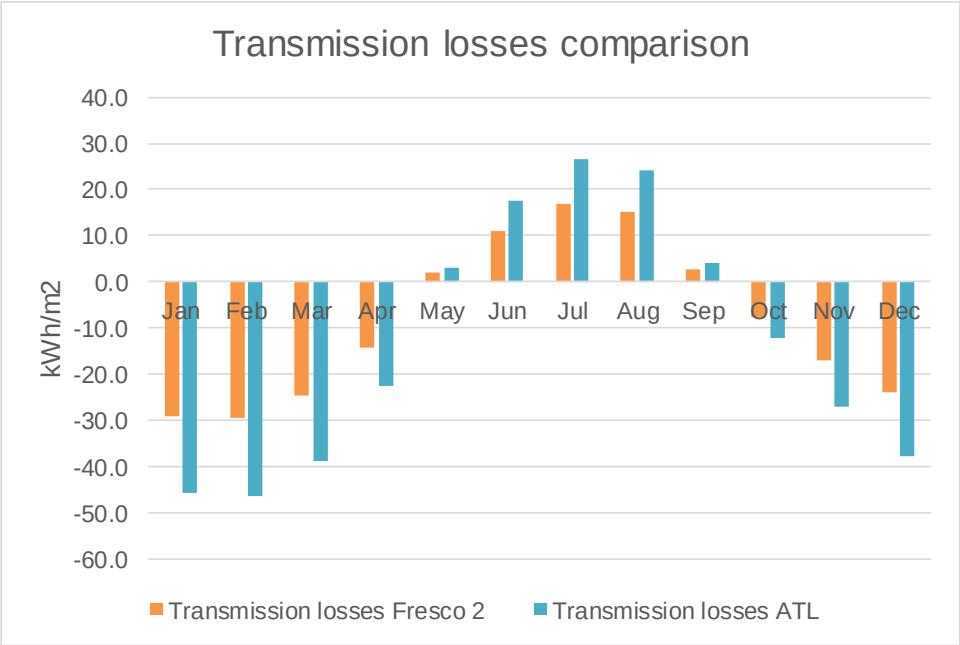


Figure 15 Comparison between the transmission losses of the ATL and Fresco 2

Ventilation losses

Since in the ATL the ventilation units are the Fresco 2 ones, there are no difference between them.

5.2.2 Internal gains

Person heat gain

The occupancy profile for the Fresco 2 is clearly the same to the ATL, since it has been taken both case from Fresco 2. This has been done in order to reduce the complexity of the model, without affect its robustness.

Solar heat gain

As it has been stated previously in the report, the geometry for every case study is the same. In the Fresco 2, selective films have been applied, because Fresco 2 could show this item. This implies that the actual radiation through the windows will be drastically reduce from 75% to 30% (25).

In the following figure the solar heat gains of ATL and Fresco 2 are compared.

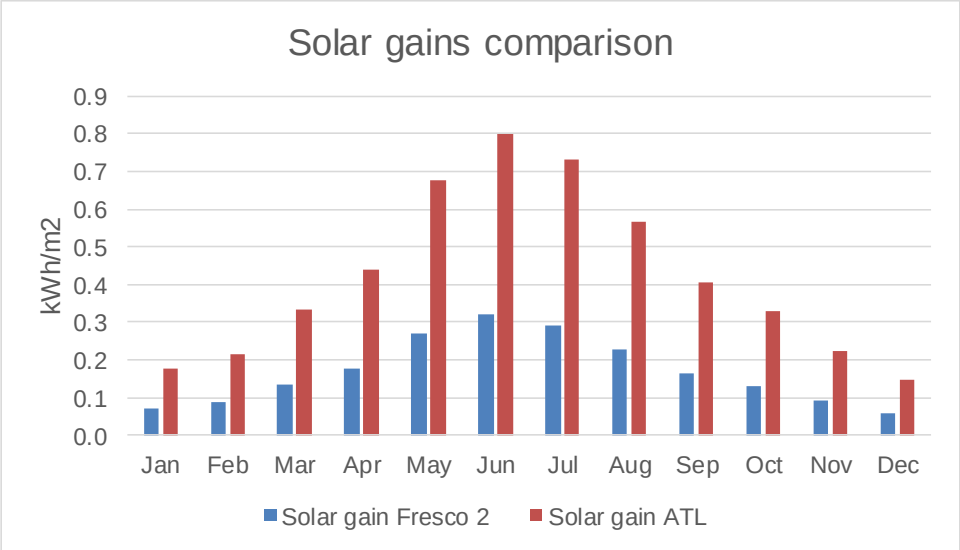


Figure 16 Comparison between the solar heat gains of the ATL and Fresco 2

5.2.3 Heat balance

The heat balance considers all the thermal losses and gains mentioned above, including also the gains due to de-humidification of internal and external moisture. Figure 17 reports the final heat balance for the Fresco 2 compared to the ATL one.

The below figure highlights that Fresco 2 performs better than the ATL, namely than the common retails in Azerbaijan. Along the entire year, the Fresco 2 case study will need less energy to satisfy the demand since the losses, both heat and cold losses, are far lower than the ATL. This means that Fresco 2 is already a good starting point for the study but large

improvements are still possible for the new acquisitions. This is what will be investigated in the REM case.

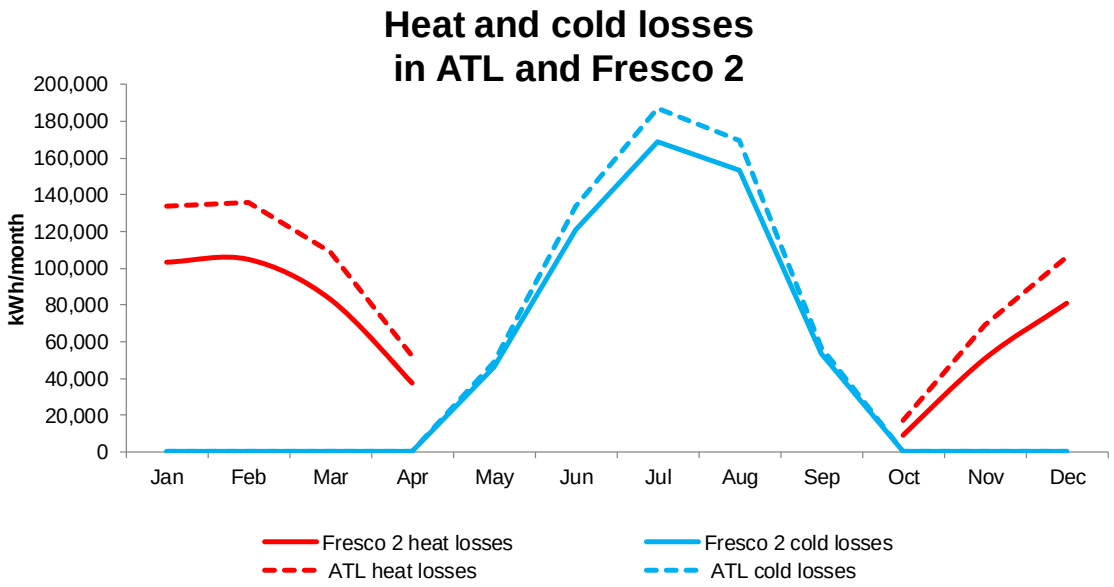


Figure 17 Heat and cold losses in the ATL and Fresco 2 case studies

5.2.4 Fresco 2 energy consumption

In this paragraph, the energy needs for matching the heat balance will be investigated.

To meet the heating and cooling needs, both VRF and rooftop units contribute.

Regarding ventilation, both types of units will be used. In particular, the main difference with the ATL case study is that the overall process has a greater efficiency. Indeed, the SFC here is 0.65 Wh/m^3 (from the catalogue) and this is an evidence that the technologies in Fresco 2 are above the average technological level in Azerbaijan. The ventilation needs of course do not change from one case study to another one, since they don't depend on the machines involved. What actually changes is the electricity consumption spent for achieving these ventilation needs.

As already explained in the report, the lighting consumptions has been calculated directly from the data gathered during the energy audit. The total power installed is 25.11 kW (since there are 584 lamps with 43 W as capacity), considering 15 hours per day of operating schedule (since the store is opened from 8:00 – 23:00) and a contemporaneity factor of 90%, the lighting consumptions is 123,400 kWh/year.

For the refrigerators, the reasoning is similar. The data has been given and presented in a dedicated worksheet of the model, the total capacity is 201.68 kW.

To conclude, the HWS consumption is equal to the ATL, since the technology involved (boiler) and the needs are the same.

As in the ATL, here the overall energy consumption is matched through electricity, therefore the primary energy conversion was needed. At this point is useful to compare it with the ATL primary energy consumption in a bar chart.

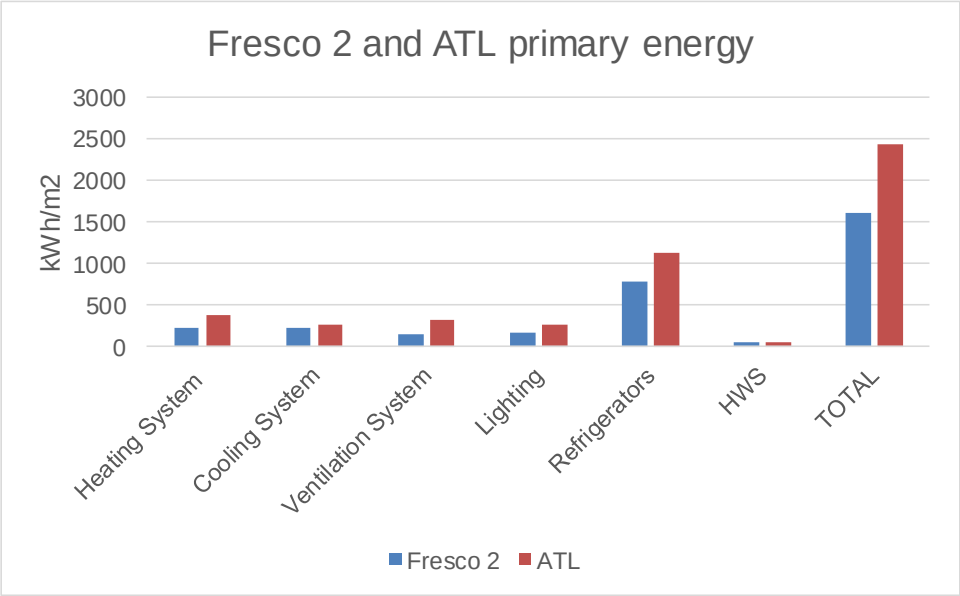


Figure 18 Comparison between Fresco 2 and ATL primary energy

To conclude, the end-use breakdown is presented.

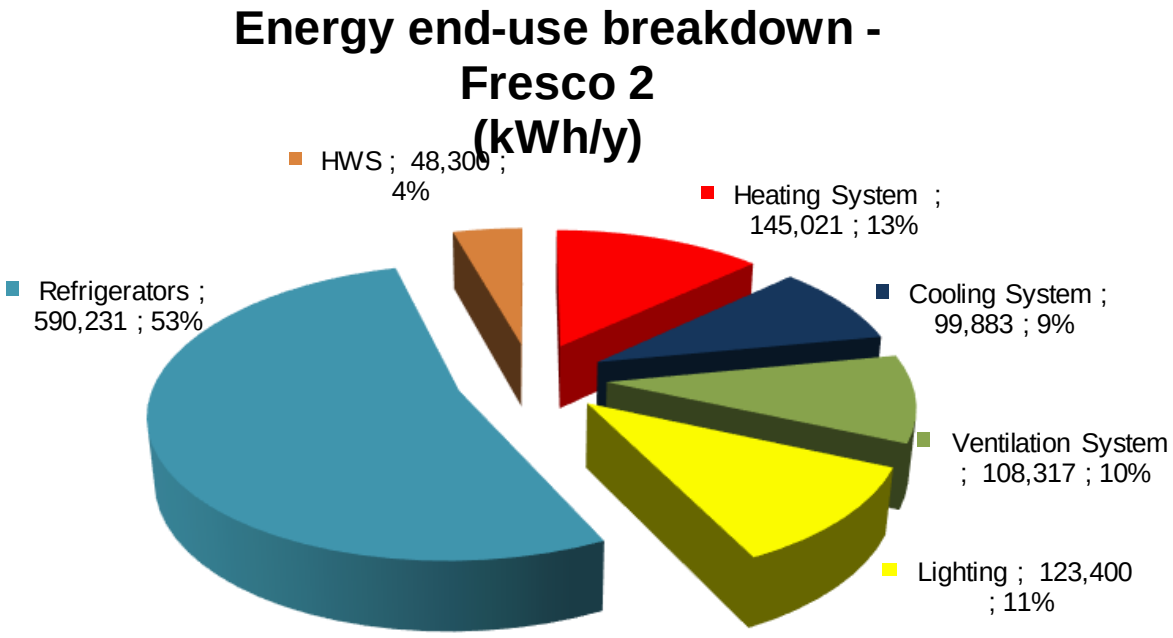


Figure 19 Energy end-use breakdown of Fresco 2

As already done for the ATL case study, also the Fresco 2 case is compared with the benchmarks retrieved and the ATL itself.

Table 8 Benchmarking of the Fresco 2 energy consumption value

Building	***Final energy use (kWh/m^2 year)***
International Benchmark (Canadian, USA) (24)	800
Eastern Europe Benchmark	890
Common practice in EU	483
Best practice	390
ATL	1,004
Fresco 2	610

This comparison gives a very important result: the current Fresco 2 state is good if compared to the ATL or the common practice of Eastern Europe Benchmarks, but still inefficient if compared to common practice in Europe. This is the reason why there is the need to propose some measures to improve the performance of this case study.

5.3 REM case study

After the Fresco 2 energy performance assessment, a list of REMs is proposed to the company for further improving the use of renewable energy resources and to increase the overall energy efficiency of the Fresco 2 level.

In the following, all measures and related investment will be evaluated considering as starting point the Fresco 2 case study, in order to show incremental benefits and costs of REMs additional investments. This means that the impact of each measure will be evaluated starting from the existing state of the Fresco 2. At the end, when the final energy balance is presented, every measure is considered at the same time and the overall saving as well. All CAPEX estimations have been made are based on data provided by the supplier, in the case of refrigerators, freezers and cold storage rooms.

To perform an economic analysis, some specific parameters have to be set. Most of them have been set by EBRD for the purpose of the project, others have been set during the thesis work. They are listed in Table 9.

Table 9 Parameters used for performing the economic analysis

Parameter	***Value***
Electricity price (EUR/MWh)	54.5
Electricity price annual escalation (%)	1
Discount factor (%)	8
Project life time (years)	15

In the following sub-chapters, a detailed description of the different REMs has been given.

5.3.1 Envelope insulation

Technical description

The Fresco 2 case study considers double glazing windows and the walls and roof do not include any specific kind of thermal insulation, but only ticker construction materials with respect to the ATL. In the REM case is suggested to enhance the insulation of the walls by realising the external wall surface structure with 25 cm brick + gypsum board + plaster + 10 cm rock wool. The external wall transmittance estimated is equal to 0.28 W/m^2K. The associated external wall surface amounts to around 710 m^2.

Furthermore, it is also suggested to improve the insulation of the ground floor and of the roof (about 2,150 m^2 for each) by installing 15 cm of expanded polystyrene (EPS). The new roof transmittance is equal to 0.22 W/m^2K and the floor estimated transmittance is equal to 0.22 W/m^2K.

Concerning the windows and glazing parts, the insulation can be improved by installing of new double glazing windows with selective film and filling them with gas (thickness 20 cm, air and argon 90%). This typology of window has U-value equal to 1.1 W/m^{2}K. The related windows' surface amounts to around 74 m^2. In the next section the energy saving obtained are illustrated, but in Table 10 the differences in U-values in the two case studies are summarised.

Table 10 U-values comparison between Fresco 2 and REM cases

Element	***Fresco 2 U-value (W/m^{2}K)***	***REM U-value (W/m^{2}K)***
Window	2.70	1.10
Wall	0.80	0.28
Roof	1.20	0.22
Floor	1.30	0.22

Resource's saving obtained

The envelope now shows better insulation properties. Indeed, the heating and cooling losses dropped drastically. This implies that also the energy consumption to meet these losses will decrease, creating an economic revenue and environmental benefits. The yearly energy (electricity) saving, coming from the thermal needs of the building, is 88.5 kWh/m^2 year, more in details 68.2 kWh/m^2 year for heating and 20.3 kWh/m^2 year for cooling. The consequent GHGs reduction is 103.6 ton/year. This is summarised in Table 11.

Table 11 Saving achieved in the REM case study thanks to the refurbishment of the envelope

Saving	***Value***
Electricity saving (kWh/m^2 year)	88.5
GHG saving (tonCO2/year)	103.6

Cash flow analysis

The additional cost of the insulation for walls, roof and floor as well as the costs of windows and doors are detailed below:

- walls insulation: 10,658 EUR (about additional 15 EUR/m2 for 10 cm thickness rock wool insulation);
- roof insulation: 32,250 EUR (about additional 15 EUR/m2 for 10 cm thickness rock wool insulation);
- floor insulation: 21,500 EUR (about additional 10 EUR/m2 for polyurethane panels);
- windows (double glazing): 4,410 EUR (about additional 60 EUR/m2).

The total additional investment cost amounts to 68,817 EUR.

Considering the electricity saving, the related cost per kWh, the discount factor and the electricity price escalation, it is possible to build a discounted cumulated cash flow analysis

and calculate the simple payback time, discounted payback time, internal rate of return (IRR) and net present value (NPV) at the end of the 15 years.

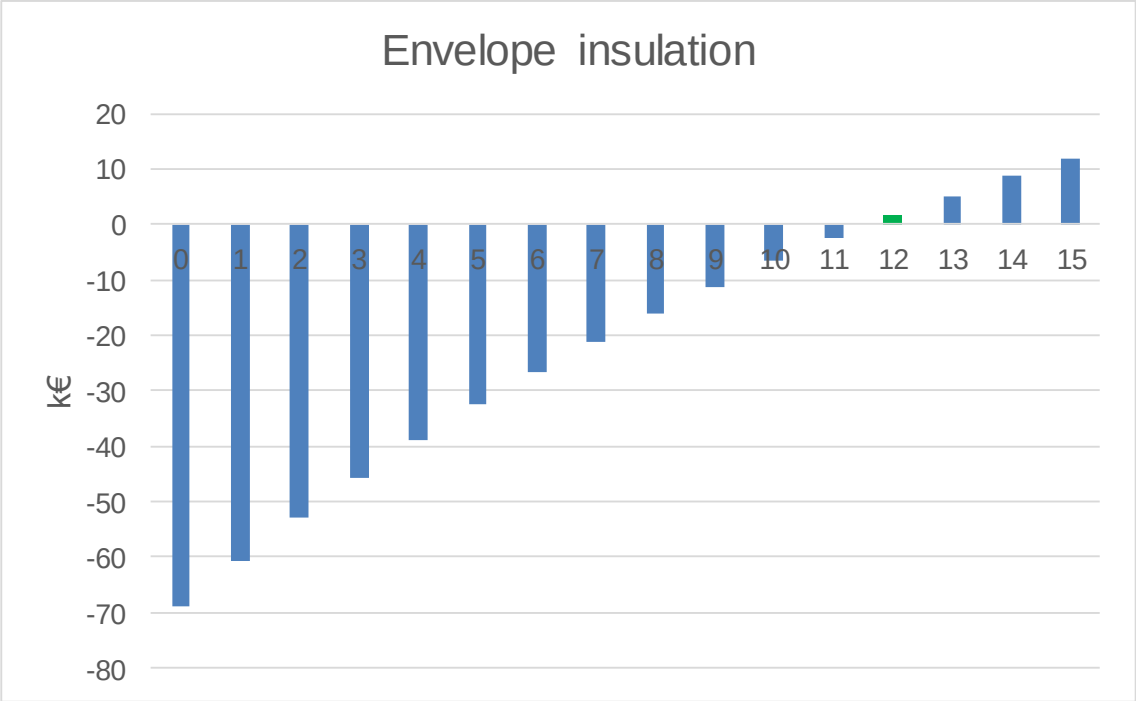


Figure 20 Envelope insulation cash flow

More in details, the simple payback time is 7.47 years, and the discounted payback time is 11.61 years. The IRR is 10.66% and the NPV is 11,918 EUR at the end of the 15 years considered. This confirms the power of the refurbishment of the envelope: not only it has a great impact on the future energy bills, but also, considering local prices, it is an investment that pays back in only 11 years.

5.3.2 Heat recovery from exhaust air

Technical description

The use of high efficiency heating ventilation and air conditioning systems was observed during the audit in Fresco 2, but no heat recovery on incoming fresh air / exhaust air has been found. This section evaluates the heat recovery potential on ventilation subsystem of Fresco 2. The proposed system is a DX type low resistance heat recovery unit with a maximum capacity of 25,000 m3/h, and average heat recovery efficiency of about 60%, to be installed between the exhaust and intake points of the existing ventilation duct (26). The DX System is a range of smart heat recovery ventilation systems that continually supply pollen-free and dust-free preheated fresh air to living areas, ensuring a comfortable feel-good climate. Using the *DynamiX® Technology*, which consists of a smart airflow control management, the DX System offers optimal indoor air quality and limits all inherent noises from the ventilation system. This technology allows to the DX System to go beyond standard energy recovery, by adapting with accuracy the airflows to the occupants needs in all rooms.

Resource's saving obtained

The heat exchanger installation would lead to a decrease of ventilation losses and consequent energy reduction of the future retails. This saving is investigated in detail in this sub-section. In the REM case, the heat contained in the exhaust air can be used to pre-heat the outside air coming in to the building. Thanks to this measure. In total, 76.4 kWh/m^2 year can be recovered thanks to this item.

In Table 12, there is the summary of the energy saved and GHGs reduction thanks to heat recovery installation.

Table 12 Saving achieved in the REM case study thanks to the installation of heat recovery HEX

Saving	***Value***
Electricity saving (kWh/m^2 year)	76.4
GHG saving (tonCO2/year)	29

Cash flow analysis

The heat recovery system chosen is *Aldag AHRV 5000*. The cost of the investment is 10,000 EUR (26).

Considering the electricity saving, and the related cost per kWh, the discount factor and the electricity price escalation, it is possible to build a discounted cumulated cash flow analysis and calculate the simple payback time, discounted payback time, internal rate of return (IRR) and net present value (NPV) at the end of the 15 years.

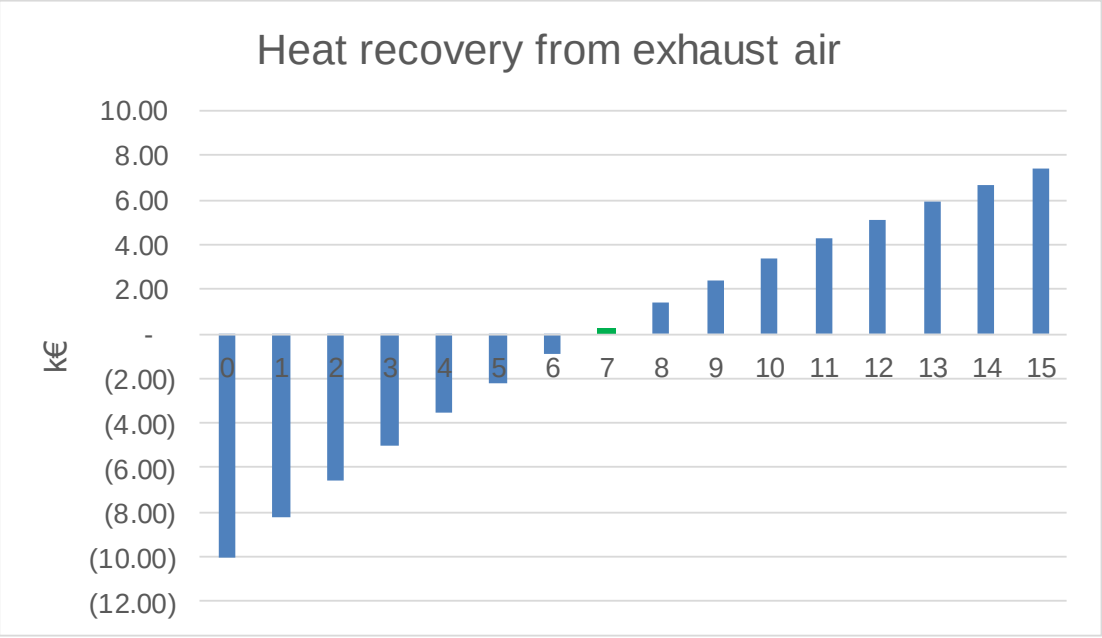


Figure 21 Heat exchanger cash flow

More in details, the simple payback time is 5.11 years, and the discounted payback time is 6.78 years. The IRR is 18.41% and the NPV is 7,369 EUR at the end of the 15 years considered. The payback period is very short, and this type of item is constantly used in several types of buildings in order to reduce the ventilation losses during the year. The temperature difference between indoor and outdoor air is always present, and through this analysis, it has been possible to highlight the potential in this country along the year. First, the investment is not very high compared to other major energy efficiency measures usually proposed, and secondly, due to the great impact on the energy consumption, the payback is short.

5.3.3 Solar tubes and LED bulbs

Technical description

Solar tubes are physical structures used for transporting or distributing natural light for the purpose of illumination. Solar light pipes, compared to conventional skylights and other windows, offer better heat insulation properties and more flexibility for use in inner rooms, but less visual contact with the external environment. The use of this equipment enables use of natural light for illuminating areas away from windows, while minimising the solar heat gain (compared to solar light coming into the building through windows). The energy savings are realised through the use of daylight sensors that turn off the electrical lighting when the daylight provided by solar tubes is enough for the lighting requirements of the area in question. As an example of this type of application Solatube daylighting products are selected. These are one of the established brands in the market. A picture of Solatube and a view of a warehouse using daylight tubes are given in Figure 22. It should be noted that the application of this measure is limited by the available roof space and the structure of the roof, as well as the distance of the internal ceiling from the roof. The applicability of the measure needs to be checked for each specific store. In any case, during discussion with Fresco Retail management, it was declared that the installation of these devices should be normally possible on the new acquisitions.

Furthermore, LED tubular bulbs have been considered in order to provide light when the solar tubes don’t work. The efficacy of the LED technologies is higher than the others. Furthermore, due to lower costs and longer lifetime, LED bulbs are becoming the most used technology in the lighting sector. An efficacy of 172 lm/W (27) has been considered, leading to a reduced power installed in the building.



Figure 22 Example of Solatube and warehouse illuminated by solar tubes

Resource's saving obtained

In the REM case study, the solar tubes will provide a huge amount of daylight. It is assumed that out of 15 hours/day (opening hours of Azeri shops), they can provide enough light for 7 hours/day. It means the lighting system should work only for the remaining 8 hours. LED power installed amounts to 14.6 kW (instead of 25.11 kW in the Fresco 2 case) due to the higher efficacy. Combining the solar tubes and the effect of having LED lights instead of TFL, the electricity consumption drops to 38,263 kWh/year from 123,400 kWh/year (Fresco 2 case), meaning an energy consumption cut of 69%.

Table 13 Saving achieved in the REM case study thanks to the installation of solar tubes

Saving	***Value***
Electricity saving (kWh/m^2 year)	46.5
GHG saving (tonCO2/year)	71

Cash flow analysis

The solar tubes technology chosen is *Solatube Skyvault* series. The investment is 38,595 EUR. Concerning LED bulbs, 18W lamp has been picked and an average cost of 9 EUR per unit leads to an investment of 7,300 EUR. In total the investment for this REM is 45,895 EUR.

Considering the resource saving coming from this measure and the electricity price, it is possible to build a discounted cumulated cash flow analysis and calculate the simple payback time, discounted payback time, internal rate of return (IRR) and net present value (NPV) at the end of the 15 years.

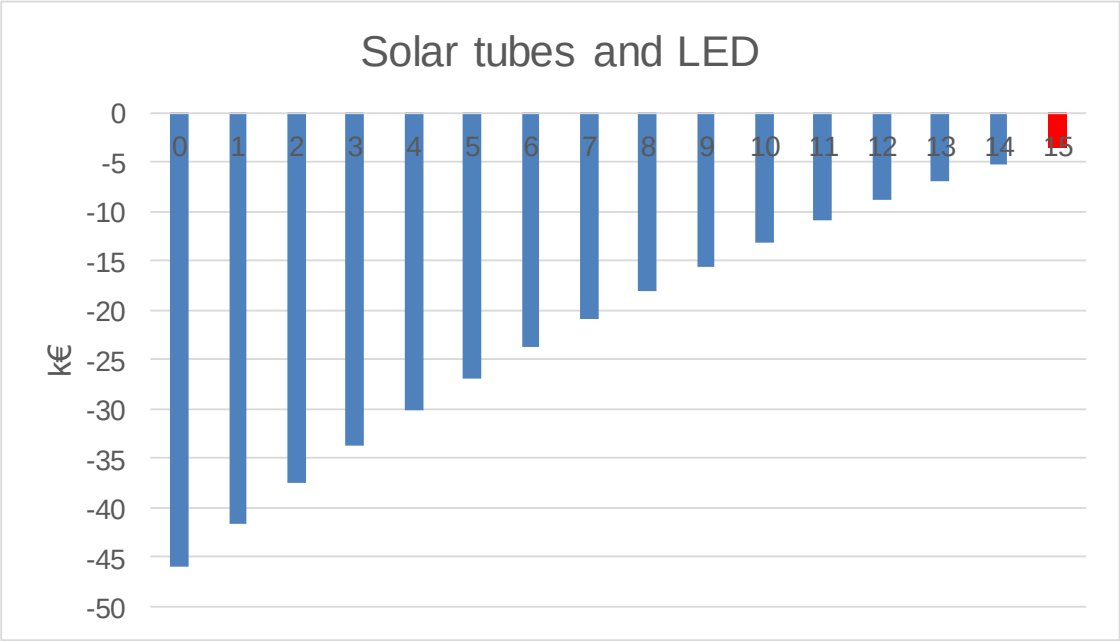


Figure 23 Solar tubes and LED cash flow

In the above figure, the negative outcome of the economic analysis can be noted. Performing a discounted NPV analysis, the discounted payback time can't be found within the life cycle limits (15 years). Instead, the simple payback time is 9.4 years, the IRR 6.79% and the NPV is -3,413 EUR. Broadening the life cycle, the discounted payback period is 17 years.
Even though the economic analysis showed a negative NPV, it is important to highlight that this technology could provide natural light avoiding energy consumption for lighting purposes and the main cause of the negative result can be the large investments. In this case, the Solatube is the only option taken in consideration. The purpose was to demonstrate that from the energy point of view, this type of technology is valuable. In this case, the economy doesn't sustain a positive and final assessment about this measure. It means that considering other options in the available market with lower prices, this option could make much more sense.

5.3.4 Double air curtains refrigerators and insulation in cold rooms

Technical description

Concerning positive temperature refrigerators, there are generally two ways to reduce energy consumption from open refrigerated display cabinets (ORDC):

- Closed cabinets (glass door or lid between customer and product);
- Double air curtains (no barrier between the customer and the product).

The advantage of open fronted vertical cabinets, and the reason why they are the preference for high-value products in supermarkets, is that there is no barrier between the customer and the product. Customers can browse and handle the products without opening the doors.

Customers can be drawn to products that they were not considering buying ("impulse buying") which may not be the case if the food is behind a door.

The advantage of the double air curtain on ORDC is that it provides an artificial barrier between the warm side of supermarket environment and cold side display area, while still allowing the customers to pick up food displayed on the shelf without physical barriers. Significant research (28) (29) showed that by imposing an artificial velocity gradient with two or more air curtains of different velocities to have the effect of smoothing out the velocity profile.

Currently, in Azeri Retail no refrigerator is equipped with air curtains. In the REM, the installation of refrigerator with air curtains is proposed. This measure is applied to the refrigerators currently present in Fresco 2. This assumption would like to show the energy improvements in the new stores to be built with the same number of refrigerators as Fresco 2; it should be highlighted that the capacity (in kW) of each refrigerator will increase of about 20%, due to the additional fans for the air curtains.

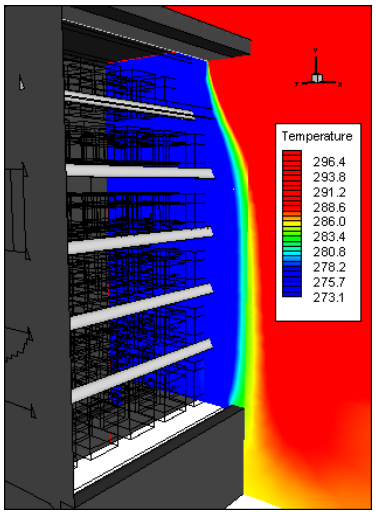


Figure 24 Example of Double Air Curtains (29)

Currently, the cold storage rooms are insulated by an insulating panel surface for a thickness of 8 cm. In the REM case is suggested enhancing the insulation of the cold storage rooms (about 88 m^2 including wall, roof and ground floor) by installing an additional insulated panel surface (thickness 8 cm).

The low temperature refrigerators and compressor units are very modern equipment that provide high performance and are built according to high quality standards. For this reason, no additional investment (compared to Fresco 2 state) for low temperature refrigerators is proposed.

Resource's saving obtained

In order to evaluate the total refrigeration load of the cold room, a simplified estimation of the heat loads has been performed:

- transmission load, which is heat transferred into the refrigerated space through its surface;
- product load, which is heat removed from and produced by products brought into and kept in the refrigerated space;
- infiltration air load, which is heat gain associated with air entering the refrigerated space.
- The installation of double air curtain refrigerators saves up to 20 % of energy with respect to installation of refrigerators without air curtains (30).

Thanks to the installation of additional insulated panel, U-value of the wall of the cold room decreases from 0.22 W/m^{2}K to 0.65 W/m^{2}K (considering EPS as insulating material). Same reasoning for roof and floor U-values, improving to 0.52 W/m^{2}K (currently is 1.8) and 0.5 W/m^{2}K (currently 1.3) respectively.

The specific heat of food and foodstuff has been taken into account for estimating the product load (31).

The total electricity saving achievable through this measure amounts to about 140 MWh/yr, 115,391 kWh/year from the double air curtains installation and 24,232 kWh/year from insulation of cold rooms respectively. The related total GHGs saving is estimated as equal to about 116 ton/yr.

The Table 14 summarises the achievable energy and GHGs savings:

Table 14 Saving achieved in the REM case study thanks to the installation of double air curtains and insulation

Saving	***Value***
Electricity saving (kWh/m^2 year)	76.4
GHG saving (tonCO2/year)	116

Cash flow analysis

In this analysis, two core components are analysed:

- Insulation in the cold rooms walls, insulation in the cold rooms roof and insulation in the cold rooms floor: the price for 8 cm of such insulating panel is 38 EUR/m^2
- Refrigerators with double air curtains: due to the lack of information available online, the cost for such refrigerators has been estimated as 20% higher than the current price of the refrigerators in Fresco 2. This is a rough estimation based on the fact that through these refrigerators is possible to achieve 20% of energy saving.

The final investment (additional cost compared to the Fresco 2 state) is 12,450 EUR for the double air curtain refrigerators and 3,344 EUR for the insulation. The discounted cash flow analysis is performed in the below figure.

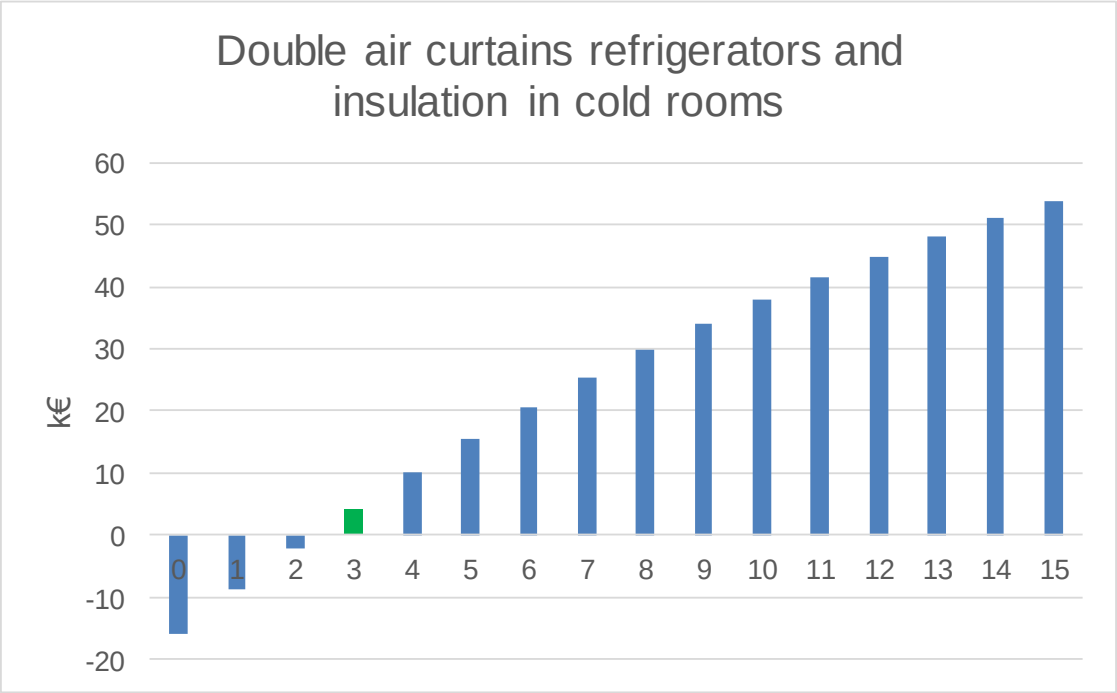


Figure 25 Double air curtains refrigerators and insulation in cold rooms cash flow

In the above figure, it is possible to realize the positive impact that this measure would have on the new built retails. More in details, the simple payback time is 2.04 years, and the discounted payback time is 2.32 years. The IRR is 49.57% and the NPV is 53,875 EUR at the end of the 15 years considered.

However, it is important to remember that the prices have been estimated in a rough way, but it is reasonable to think that they are not so far from the actual values. Therefore, the outcome of this analysis doesn't lose its value.

5.3.5 REM heat balance and energy consumption

In this paragraph, every measure above described is considered at the same time. This means that the heat balance and consumption of the building is referred to a new building with different but better characteristics than Fresco 2.

The energy consumption of this case study is expected to be lower than the other two. This is because of additional energy saving measures taken starting from the Fresco 2 (current Fresco 2 state).

Before analysing the improvements achieved in the energy consumption, it is very important to relate the consumption to the thermal losses. Indeed, if the thermal losses of a building are reduced, the energy consumption is supposed to decrease as well.

In Figure 26 the thermal losses along the year in the REM case are compared to the ones in the Fresco 2 case and depicted in one single chart.

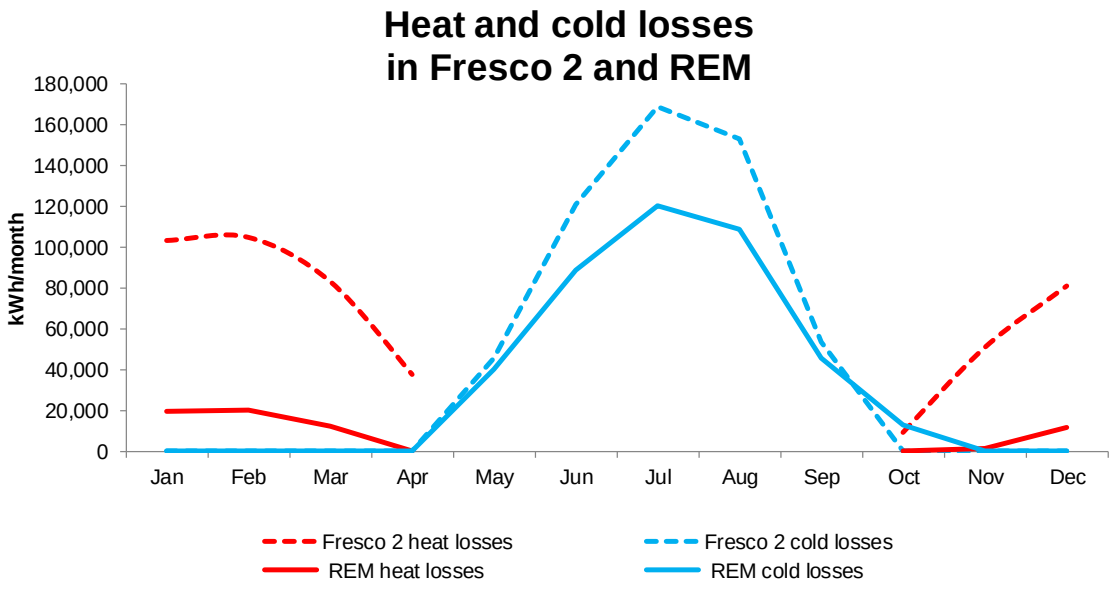


Figure 26 Heat and cold losses in the Fresco 2 and REM case study

As can be seen from the chart, in the REM case the building is able to store in a better way the indoor temperature, since the thermal losses are drastically reduced. Focusing on heat losses, thanks mainly to envelope and waste heat recovery heat exchanger, in January only the transmission and ventilation losses have been reduced from 129 MWh to 44 MWh. This is an excellent result since the heating consumption, after the refrigerator sector, accounted for the largest share in the end-use breakdown of the Fresco 2 (13%). In the warmest months the cold losses drop as well, meaning that the resource efficiency of the building increased and it brings something positive and valuable for Azeri Retail's future supermarket.

Furthermore, the energy consumption is the next step.

Having the building losses, the technologies' characteristics, calculating the energy consumption is straightforward. In the below figure, there is a comparison between the primary energy use in the two last case studies.

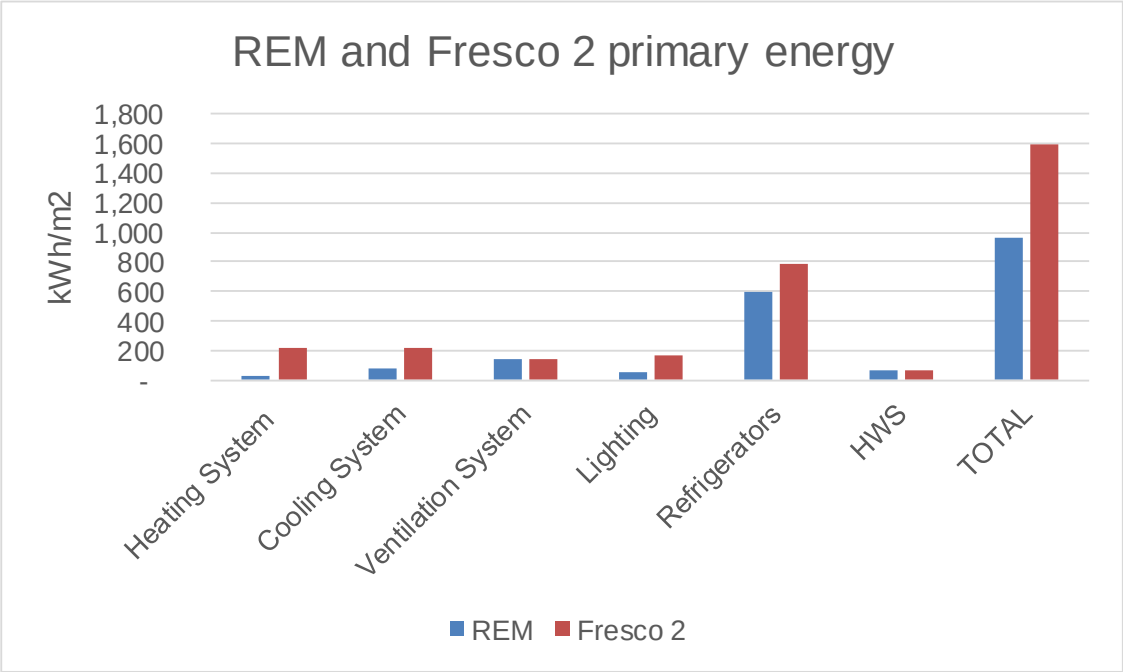


Figure 27 Comparison between REM and Fresco 2 primary energy

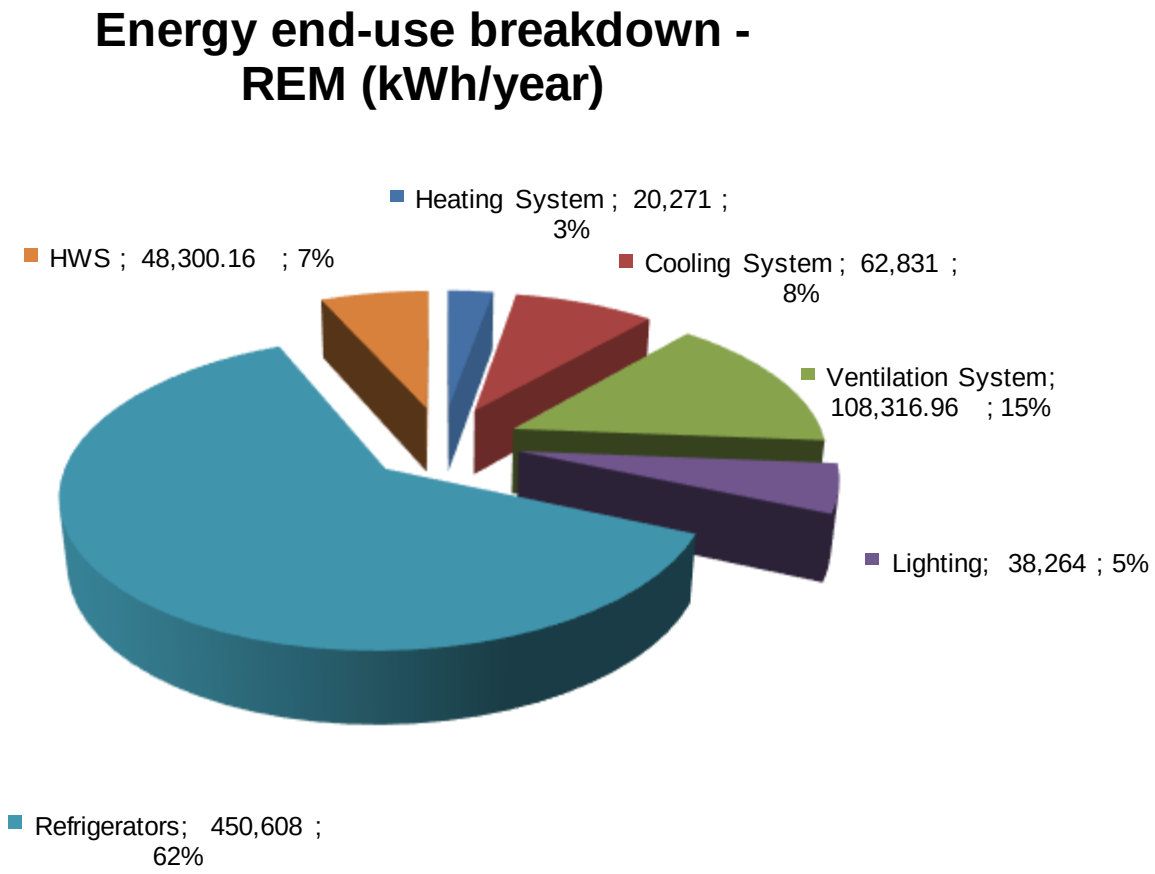


Figure 28 Energy end-use breakdown of the REM

Table 15 Benchmarking of the REM energy consumption value

Building	***Final energy use (kWh/m^2 year)***
International Benchmark (Canadian, USA) (24)	800
Eastern Europe Benchmark	890
Common practice in EU	483
Best practice	390
ATL	1,004
Fresco 2	610
REM	399

6 RESULTS

In this thesis project, an energy audit has been conducted in Azeri Retails stores in Azerbaijan. The audit consisted in a site visit, collection of interesting data, analysis of current energy performance of Fresco 2 store and further proposal of resource efficiency measures in order to make the future stores, that Azeri Retail would like to refurbish, more efficient.

Three case studies have been analyzed, each one representing something different, as clearly explained throughout this report. In the below figure they are compared in order to visualized clearly the savings achieved in terms of final energy use expressed as kWh/m^2. Together with the results achieved with the model built, it is useful to compare these with benchmarks retrieved.

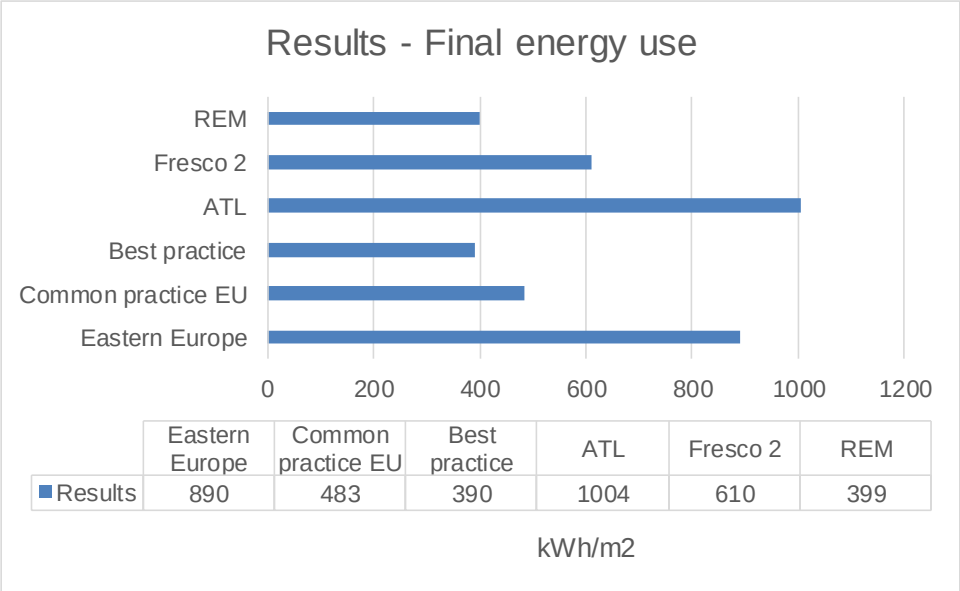


Figure 29 Final results and comparison about energy use

Looking at the above Figure, it is possible to add some interesting comments: the ATL is highly inefficient, indeed consumes more than double of the energy use commonly in European supermarkets. Furthermore, current Fresco 2, if compared to Azerbaijan's standards, can be judged positively, but looking at the common practice in EU it is clear that some improvements are possible. This is what is researched in REM case study where, through the application of four measures, the final energy use drops to 399 kWh/m^2.

As can be seen from the end-use breakdowns of the three case studies presented in the above sections, the percentages for the energy needs have not changed a lot. This is because of the specific scope of this kind of building: the refrigerators typically account for the largest share of the total energy use, and this is due to the type of the retail. Furthermore, heating system in the building sector is generally one of the largest energy-consumer sector. However, in the third case an excellent combination of insulating materials has been considered and this

is reflected in a drastic reduction of heat losses and heating consumption. However, it is still interesting the comparison between the three of them. The refrigerators, although they account for more than 46% in every case study, have lowe energy demand than in the third one, passing from 852 (ATL) to 451 MWh/year (REM). And this can be applied also to the heating system: the share is 15% in the ATL, 13% in the Fresco 2 case study and 3% in the REM case and in absolute value the cut is 263 MWh/year from ATL to REM.

Since one of the research questions clearly addresses the environmental impact of the supermarkets, it is also useful to compare the GHGs emissions that each case study releases annually.

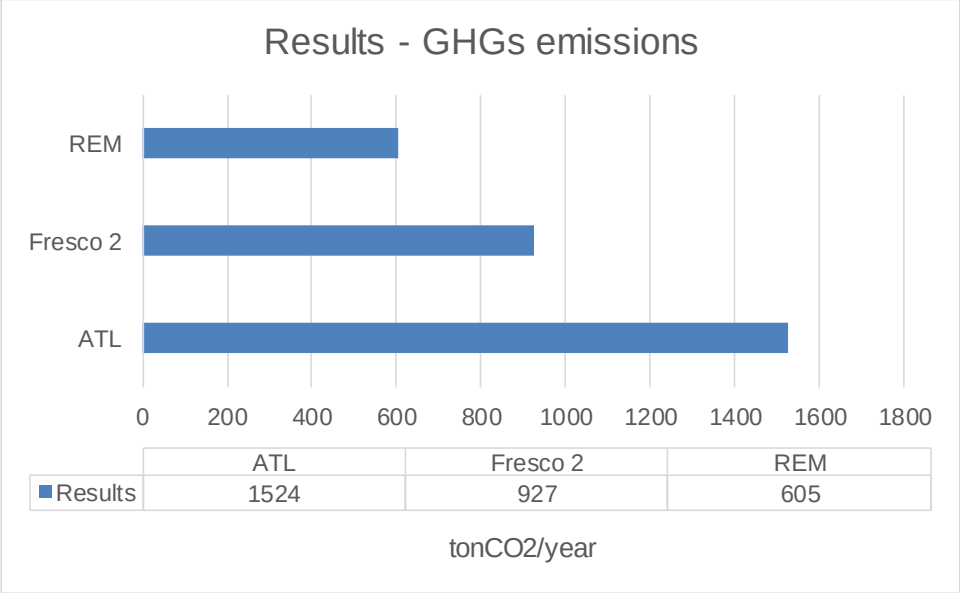


Figure 30 Final results and comparison about GHGs emissions released

A direct consequence of the energy efficiency improvement in the REM case study is the reduction in GHGs emissions released. The current state of Fresco 2 is more environmental friendly than the common practice but through the REM case study it has been proved that there are margins for improvements.

The GHGs emissions, as well as the final energy use, come directly from the technology used. This means that according to the equipment chosen and the fuel used, the energy efficiency of the building can vary a lot. It is useful to remember that in order to pass from electricity and primary energy, a PEF of 2.43 kWhprimary/kWhel, this implies that more than two kWh of fuel are necessary to generate 1 kWh of electricity. That's the reason why also the primary energy use has been investigated in the thesis.

Below, the primary energy utilization of the three case studies is presented below.

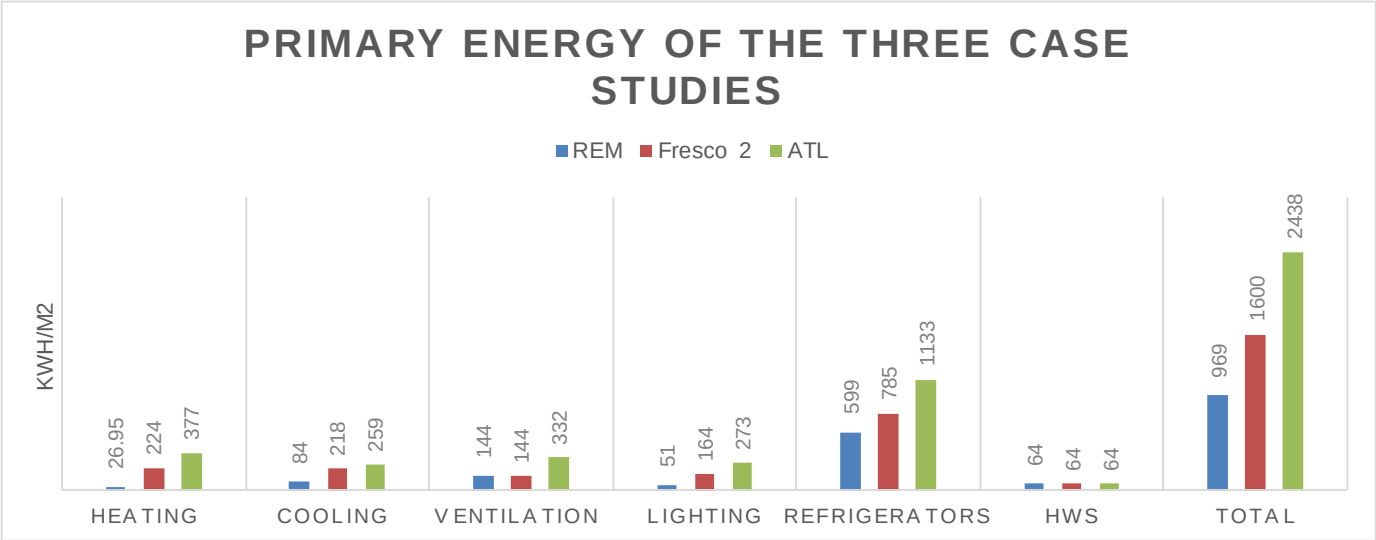


Figure 31 Primary energy use comparison between the thee case studies

In the three case studies there is no use of natural gas or other fuel or fossil fuel. This implies that the final energy for each system is smaller than the primary energy consumed and absorbed by the grid. The HWS are the same for each case study. This was driven by the fact that it is not reasonable to assume that the use of heat for warming the water decreases. Generally, the use of energy for heating the sanitary hot water is very difficult to decrease, since in most of the cases the hot water consumed is driven by natural needs (shower, toilets, kitchens) and the reduction is hard. On the other side, the installation of new lighting fixtures led to a considerable improvement: from 273 kWh/m2 to 51 kWh/m2. This means that the electricity used, following the thesis' advices, is almost 1/6 of the current. Being a supermarket with big businesses coming from the frozen products, the refrigeration sector saves 24% of the current energy use (Fresco 2 case).

One of the strong points of this research is that it achieves good results with a small investment. Indeed, summing up all the investments needed for the energy efficiency process suggested in the thesis, the final cost would be 140,000 EUR. The good point is that at the end of the 15 years, the NPV is definitively positive. This is shown in the below picture.

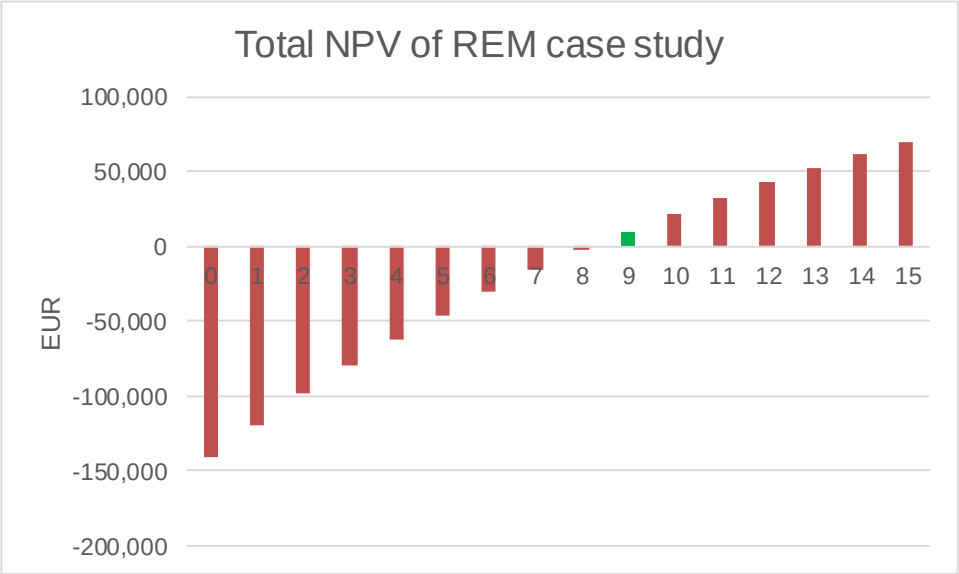


Figure 32 Total Net Present Value of the REM case study

Going deeper in the analysis of the NPV chart, it is possible to see that the discounted PB is basically only 8 years. This short period of time is totally acceptable from the supermarket's perspective. Indeed, big facilities are used to make complex business models and business plans because the investments are huge and the payback is important to be not too far from the "year 0". In this case, the thesis' aim was to suggest low cost but effective measures, able to decrease the carbon footprint of the supermarket, having a good payback and NPV at the end of 15 years.

The thesis tries to answer to the research questions mentioned at the beginning of this report. Through the ATL case, the current energy efficiency state of supermarkets in Azerbaijan has been investigated. Typically, the envelope doesn't show insulating properties: this highly affects the thermal losses of the structure. Furthermore, the refrigerators, representing the larger share in the total energy consumption of the building, are not equipped with air curtains. As it has been shown in the REM case, this measure can drastically reduce the energy use. It is difficult to answer what is the most competitive energy efficiency measure for supermarket in Azerbaijan. Through the benchmarking, several cases have been compared to the three developed in this thesis and this process gives more robustness and more value to the overall analysis carried out. Each value has been normalized (expressed as kWh/m2) for this reason.

7 DISCUSSION AND CONCLUSIONS

In general, even though the great majority of the buildings present common characteristics, each of them is different due to the technology installed, their regulations and occupants. Also, in order to be able to compare them it is essential to try to establish a methodology and tools supporting the energy audit. Besides, the validation of a methodology is based on the analyses of collected data and the established models but these ones are the results of hypothesis or approximations. An energy audit can only offer an overview of the situation of a building at a specific moment. At the same time, the choice of the software is also a source of errors because the simulations of the same building with the same input data but with two different energy modeling software typically does not lead to the exact same result. As a result, the energy audit is a precious tool based on important source of errors. Nonetheless, due to the balance between environmental policies and businesses it offers the possibility to establish the energy breakdown of a building.

In this case, the consultancy company hadn't plan to use a building modeling software, that's the reason why my major task in the project has been building a model with *Microsoft Excel* in order to have the most accurate possible analysis with the available tools. Obviously, with the aid of a proper building modeling software it would be possible to achieve more accurate values and explore different measures to be proposed.

Therefore, future work is recommended to confirm and further improve the results achieved. More complex building simulation software, calculations for HVAC measures with hourly simulations, longer site visits and measurements are recommended. Obviously, in order to do this, more time, more resources are needed and those recommendations were out of the thesis scope. However, it is reasonable to believe to the results achieved in this thesis project because all the methodology is robust, reliable and valid due to the rigid and scientific approach conducted throughout the thesis period.

One of the strong points of this research is that it achieves good results with a small investment. Indeed, summing up all the investments needed for the energy efficiency process suggested in the thesis, the final cost would be 140,000 EUR. The good point is that at the end of the 15 years, the NPV considering all the measures proposed is definitively positive, as shown in the picture above. This analysis showed that a considerable amount of energy can be saved in a supermarket facility. Of course, the energy saved largely depends on the investment that the owner of the supermarket is willing to spend. In this case, the analysis tried to focus on low cost measure that could help the building to reduce its carbon footprint. Summing up all the investments required for the four measures, 140,000 EUR is not a very high amount of money, but with this expense, the owner can save up to 605 kWh/m^2 of final energy use every year.

The thesis tries to answer to the research questions mentioned at the beginning of this report. During the analysis carried out in the ATL case study, it turns out that the current energy efficiency state of supermarkets in Azerbaijan is very poor. Typically, the envelope doesn't show insulating properties and no heat exchangers to preheat or precool the incoming air are present in the low-quality supermarkets and this highly affects the thermal losses of the entire building structure. Furthermore, the refrigerators, representing the larger share in the total energy consumption of the such supermarkets, are not equipped with air curtains. The fourth measure has shown that the installation of such air curtains can drastically reduce the energy use.

In general, it is difficult to answer what is the most competitive energy efficiency measure for supermarket in Azerbaijan. That's the reason why in the thesis project, a set of four measures is presented that cover all the major loads of the supermarkets (based on frozen food products): refrigerators, lighting, envelope and heat losses. Anyway, if an exact answer is needed, it is possible to state that, if most of the supermarket's businesses come from frozen products, literature and this project show that refrigerators should be the major load to be taken into account in an energy efficiency perspective.

BIBLIOGRAPHY

1. **IEA.** Energy Efficiency: Buildings. [Online] https://www.iea.org/topics/energyefficiency/buildings/.

2. **UN Environment.** *GLOBAL STATUS REPORT 2017 - Towards a zero-emission, efficient, and resilient buildings and construction sector.* 2017.

3. **SKM Enviros.** randd.defra.gov.uk/Document.aspx?Document=9873_Appendix-ProjectReportv8a.pdf. [Online]

4. **Sintef.** The Future Energy Effective Supermarket. [Online] 2017. https://www.sintef.no/projectweb/creativ/the-future-energy-effective-supermarket/.

5. **R. Smith, K. Kim and J. Klemes.** Reduction of refrigeration energy consumption and environmental impacts in food retailing, Handbook of Water and.

6. **Zoi Mylona, et al.** *Comparative analysis on the energy use and environmental impact of different refrigeration systems for frzon food supermarket application.*

7. **Zoi Mylona, Maria Kolokotroni and Savvas Tassou.** *Frozen food retail: Measuring and modelling energy use and spaceenvironmental systems in an operational supermarket.*

8. **CIRCE.** *SME - Energy CheckUp.*

9. **DNV GL.** *A Study on Energy Efficiency in Enterprises: Energy Audits and Energy Management Systems.*

10. **ENERGY CHARTER SECRETARIAT.** *In-Depth Review of the Energy Efficiency Policy of Azerbaijan.*

11. **Forum, European Retail.** *Issue paper on energy efficiency stores.*

12. **MWH - Building a better world.** *Identification of energy efficiency opportunities at retail outlets in Ukraine and EE penetration rate in other neighbouring countries.*

13. **Milan Polytechnic - Dr. Alessandro Dama,Energy dep.** *Glazing and technologies for windows with high energy performance.*

14. **Giuliano Dall'O'.** *Green Energy Audit of Buildings - A Guide for a Sustainable Energy Audit of Buildings.*

15. **ABRAXAS.** Types of Energy Audits. *Commercial Energy Audits.* [Online] https://www.abraxasenergy.com/energy-consulting/commercial-energy-audits/types-of-energy-audits/.

16. **OeEB and Frankfurt school.** *Energy efficiency finance - Azerbaijan.*

17. **Energy - Models.** Internal Heat Gains (IHG). *Energy - Models.* [Online] http://energy-models.com/internal-heat-gains-ihg.

18. **Ali Bahadori-Jahromi, et al.** *Impact of Window Films on the Overall Energy.*

19. **NYC Department of Design & Construction.** *Manual for quality, energy efficient lighting.*

20. **The Canadian journal of infectious diseases.** *Residential water heater temperature.*

21. **Plumbing Manufacturers Institute.** *Water temperature control and limitation.*

22. **National Institute of Natonal Health of Quebec.** *Prevention of legionella preliferation in sanitary hot water.*

23. **EBRD .** *Electricity Emission Factors Review.*

24. **Istitut de recherche Hydro-Quebec.** *Available data from different Canadian supermarkets.*

25. **Information, Depth tutorials and.** Window Films: Spectrally Selective versus Conventional Applied (Energy Engineering). *What - When - How Energy Engineering.* [Online] http://what-when-how.com/energy-engineering/window-films-spectrally-selective-versus-conventional-applied-energy-engineering/.

26. **Aldag.** HVAC Systems. *Archiexpo.* [Online] http://pdf.archiexpo.it/pdf-en/aldag-heating-cooling-air-conditioning-ind-trd-inc/general-catalog/149510-276679.html#open.

27. **Philips.** MAS LEDtube. *http://www.lighting.philips.com/main/prof/led-lamps-and-tubes/led-tubes/master-ledtube-em-mains-t8/929001377002_EU/product.* [Online]

28. **A. M. Foster, et al.** *Three-dimensional effects of an air curtain used to restrict cold room infiltration.*

29. **Paul Amitesh and Tiwari A. C.** *A comprehensive review on design and performance of air curtains.*

30. **Freor - thinking green.** https://freor.com/freors-dual-air-curtain/. *https://freor.com/.* [Online]

31. **The Engineering Toolbox.** Engineering Toolbox - specific heat capacity of food and foodstuff. *The Engineering Toolbox.* [Online] https://www.engineeringtoolbox.com/specific-heat-capacity-food-d_295.html.